INCH-POUND

MIL-M-28787/365A
3 August 1994
SUPERSEDING
MIL-M-28787/365
26 October 1990

MILITARY SPECIFICATION

MODULES, DIGITAL, STANDARD ELECTRONIC

INTERFACE MODULE, KEY CODE DGP

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the modules described herein shall consist of this specification and the latest issue of MIL-M-28787.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic interface module. This module is class I as specified in MIL-STD-1389.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Code 603A, Bldg. 2917, NAVSURFWARCENDIV, 300 Highway 361, Crane IN 47522-5001, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification, standards, and handbook. The following specification, standards and handbook form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

HANDBOOK

MILITARY

MIL-HDBK-217 - Reliability Prediction of Electronic Equipment.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Qualification. Modules furnished under this specification shall be products which are authorized by the qualifying activity for listing on the applicable qualified products list by the time of award of contract (see 4.5).

3.2 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.3 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

3.4 Failure rate requirements. All modules procured to this specification shall have a failure rate predicted in accordance with MIL-STD-1389. The predicted failure rate shall not exceed 1.1820 failures per million operating hours when predicted in accordance with MIL-HDBK-217.

3.5 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I as specified in MIL-STD-1389.

3.6 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.6.1 Key code. The module key code shall be DGP.

3.6.2 Size. The module size shall be 2A Format B.

3.6.3 Weight. The module shall have a maximum weight of 250 grams.

3.6.4 Contact pins. The module shall have 100 contact pins positioned 1 through 50 in one row and 51 through 100 in the other row.

3.6.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.6.6 Depth. The module shall have no additional depth requirements.

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TABLE I Module electrical requirements

Test cond	Parameter 5/	Ref test table	Test type 1/ - 3/	25 ± 5°C 4/		0 ± 5°C 4/		60 ± 5°C 4/		Units
				Min	Max	Min	Max	Min	Max	
A	Isolation, initial	II	S	10	INF	10	INF	10	INF	MOhm
	end-of-life	II	S	1	INF	1	INF	1	INF	MOhm
B1-B4	Continuity	III	X	0 0	1 0	0 0	1 0	0 0	1 0	Ohm
B5-B8	Continuity	III	Q	0 0	0 1	0 0	0 1	0 0	0 1	Ohm
C	Vcc stress	IV	Q	Module shall pass all remaining tests						
D1-D31	Ibd	V	S	0	100	0	100	0	100	µA
D32,D33					10		10		10	
D34-D49	Ibd	V		0	100	0	100	0	100	µA
E1-E47	Vic	VI		-1 5	-0 4	-1 5	-0.4	-1.5	-0 4	V
F1-F32	Ios	VII		-225 0	-40.0	-225.0	-40.0	-225 0	-40 0	mA
F33-F36				-120.0		-120.0		-120.0		
F37-F46				-225 0	-40 0	-225.0	-40.0	-225 0	-40.0	
F47	Ios	VII	S	-6.3	-5 2	-6.3	-5.2	-6.3	-5.2	mA
G1-G29	Iih	VIII	X	0	20	0	20	0	20	µA
G30					40		40		40	
G31	Iih	VIII	X	0	20	0	20	0	20	µA

- See footnotes at end of table

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/ - 3/	25 ± 5°C 4/		0 ± 5°C 4/		60 ± 5°C 4/		Units
				Min	Max	Min	Max	Min	Max	
G32,G33	Iih	VIII	X	-10	10	-10	10	-10	10	μA
G34-G49	Iih	VIII		0	20	0	20	0	20	
H1-H27	III	IX		-200	-5	-200	-5	-200	-5	
H28				-1985	-1285	-1985	-1285	-1985	-1285	
H29				-780	-80	-780	-80	-780	-80	
H30,H31				-900	-200	-900	-200	-900	-200	
H32,H33				-400	-30	-400	-30	-400	-30	
H34				-1600	-700	-1600	-700	-1600	-700	
H35				-200	-5	-200	-5	-200	-5	
H36,H37				-10	10	-10	10	-10	10	
H38-H53	III	IX		-200	-5	-200	-5	-200	-5	μA
J1-J47	Voh	X		2.4	4.5	2.4	4.5	2.4	4.5	V
J48					3.0		3.0		3.0	
J49,J50	Voh	X		2.4	2.8	2.4	2.8	2.4	2.8	
K1-K52	Vol	XI		0.0	0.4	0.0	0.4	0.0	0.4	
K53	Vol	XI	X	0.26	0.4	0.26	0.4	0.26	0.4	V

- See footnotes at end of table

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* TABLE I Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/ - 3/	25 ± 5°C 4/		0 ± 5°C 4/		60 ± 5°C 4/		Units
				Min	Max	Min	Max	Min	Max	
L1-L16	Iozh	XII	X	-20	20	-20	20	-20	20	µA
L17-L20				-100	100	-100	100	-100	100	
L21	Iozh	XII		-200	200	-200	200	-200	200	
M1-M16	Iozl	XIII		-200	20	-200	20	-200	20	µA
N	Icc	XIV		35	705	35	705	35	705	mA
P1,P2	Functional	XV	X	See table XV						
Q1,Q2	Dynamic	XVI	S	See table XVI						
Q3,Q4	Dynamic	XVI	Q	See table XVI						
R	PSFC	XVIII	S	8 1	12 3	8 1	12.3	8 1	12.3	µF
S	PSTA	XIX	Q	0	400	0	400	0	400	mV
* T1-T16	Output crosstalk (high level)	XX			1100		1100		1100	
* T17-T32					750		750		750	
* T33-T54					400		400		400	
* U1-U16	Output crosstalk (low level)	XXI			1100		1100		1100	
* U17-U32					750		750		750	
* U33-U52			Q	0	400	0	400	0	400	mV

- See footnotes at end of table

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* TABLE I Module electrical requirements - Continued.

	Test cond	Parameter 5/	Ref test table	Test type 1/ - 3/	25 ± 5°C 4/		0 ± 5°C 4/		60 ± 5°C 4/		Units
					Min	Max	Min	Max	Min	Max	
*	V1-V16	Output crosstalk duration, low level	XXII	Q	0	75	0	75	0	75	nS
*	V17-V24					153		153		153	
*	V25-V32					128		128		128	
*	V33-V48	Output crosstalk duration, high level				75		75		75	
*	V49-V56					153		153		153	
*	V57-V64		XXII		0	128	0	128	0	128	nS
*	W1,W3-W8, W19-W25, W35	Cin	XXIII		2	25	2	25	2	25	pF
*	W2, W13-W18, W32-W34					35		35		35	
*	W9-W12, W26,W27, W46-W53					40		40		40	
*	W28-W31					75		75		75	
*	W36-W45	Cin	XXIII			30		30		30	
*	X1-X4	Cout	XXIV			25		25		25	
*	X5	Cout	XXIV	Q	2	50	2	50	2	50	pF

- See footnotes at end of table

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• TABLE I Module electrical requirements - Continued.

	Test cond	Parameter 5/	Ref test table	Test type 1/ - 3/	25 ± 5°C 4/		0 ± 5°C 4/		60 ± 5°C 4/		Units
					Min	Max	Min	Max	Min	Max	
•	X6	Cout	XXIV	Q	2	25	2	25	2	25	pF
•	Y1,Y2	Threshold	XXV	S	See table XXV						
•	Z1-Z16	Disable time, TPLZ	XXVI	Q	4	260	4	260	4	260	nS
•	Z17-Z32	Disable time, TPLZ				130		130		130	
•	Z33-Z48	Disable time, TPHZ				260		260		260	
•	Z49-Z64	Disable time, TPHZ	XXVI		4	130	4	130	4	130	nS
•	AA1	Noise margin	XXVII	Q	0	800	0	800	0	800	mV

- See footnotes on next page

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1/ "X" parameters shall be 100 percent tested during the module acceptance test.

2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.

3/ "Q" parameters shall be tested during qualification testing only.

4/ Unless otherwise specified in the test table, limits are for initial and end-of-life.

5/ Abbreviations used in table I are defined in 6.1.

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3.6.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.6.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part or identifying number M28787/365-1. No marking shall interfere with heat dissipation from the fin (such as affixing a label which may act as a heat barrier).

3.7 Circuit pin configuration. The functional block diagram shall conform to figure 1. The pin assignments shall conform to figure 2.

3.8 Acceptance test. All production modules shall be tested in accordance with footnotes 1 and 2 in table I.

3.9 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits in table I (see 6.2).

3.10 Power supply voltage. The module shall be capable of conforming to the requirements in table I for a power supply voltage (Vcc) of 5.0 ± 0.5 V dc, except the output short circuit current limits apply only for a power supply voltage of 5.5 V dc ± 1 percent.

3.11 Parallel outputs. Paralleling of outputs is not permissible.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. Unless otherwise specified herein, all resistors shall be metal film and all capacitors shall be ceramic.

4.4 Test procedures. The module shall be tested as specified in table I.

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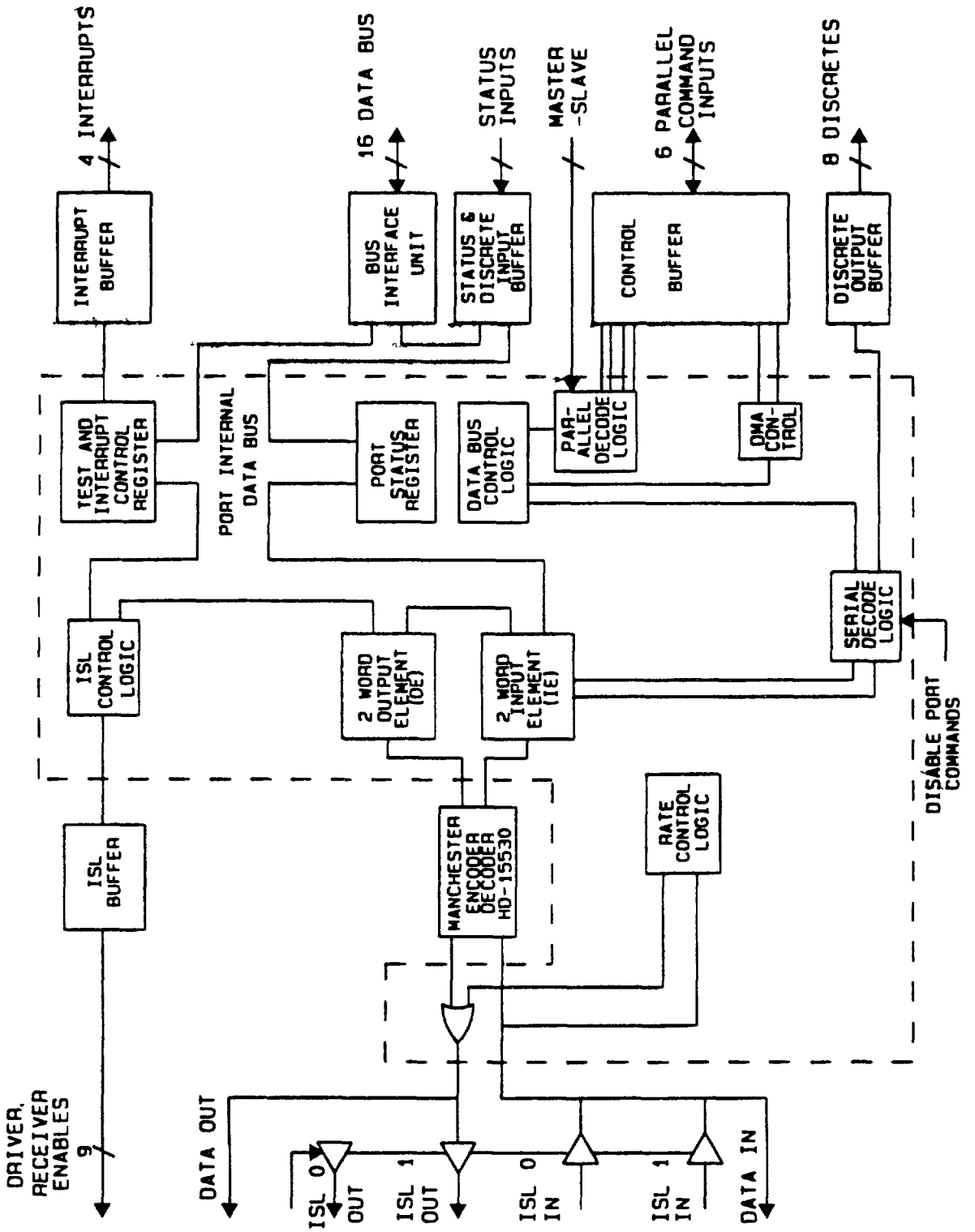


FIGURE 1. DGP functional block diagram.

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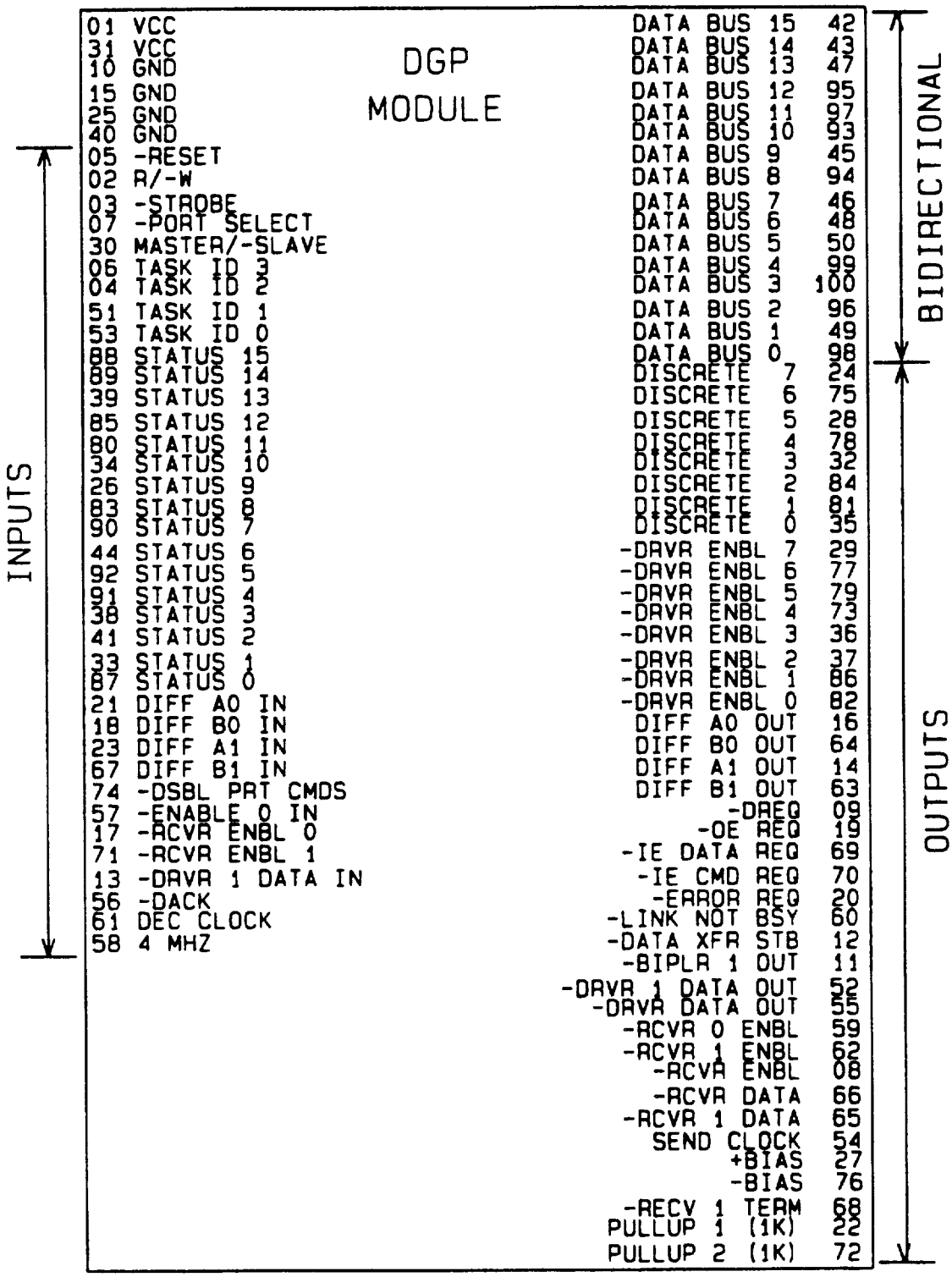


FIGURE 2. DGP pin assignment.

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4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Quality Assurance Activity (SEM-QAA) and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through Y shall be performed in any order with the following exceptions. Test conditions A, C, D, E, and F are stress tests and shall be performed only at 25 ± 5°C. If test conditions A, C, D, E, or F are part of a sequence of tests, the order of testing shall be as follows: A, C, D, E, and F where applicable, followed by the remaining tests in the sequence.

4.4.3 Multiple testing of pins. Singular pins listed multiple times in a test table indicate multiple devices connected to the pins. The module shall be tested so that each device connected to a pin is exercised and tested.

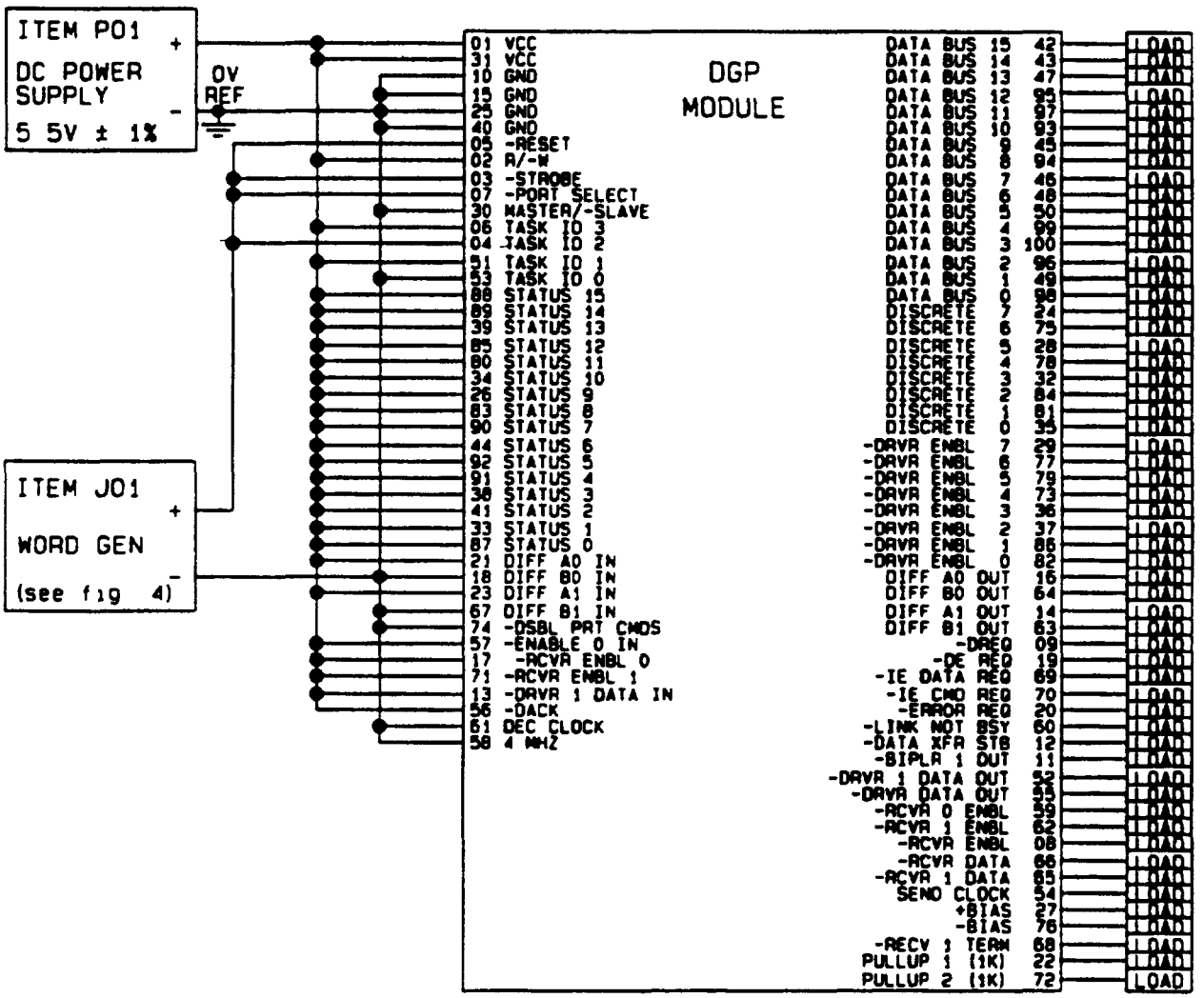
4.5 Qualification testing. Initial qualification tests and periodic check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

* 4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through Y in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions B, G through V, Y and Z shall be performed in the order specified in 4.4.2 using the initial limits for the applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated as shown on figures 3 and 4.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 5.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 6. Following the vibration tests, the electrical tests in table I which are marked with an "X" in the test type column, and test condition R, shall be performed using the initial limits at 25 ± 5°C ambient.

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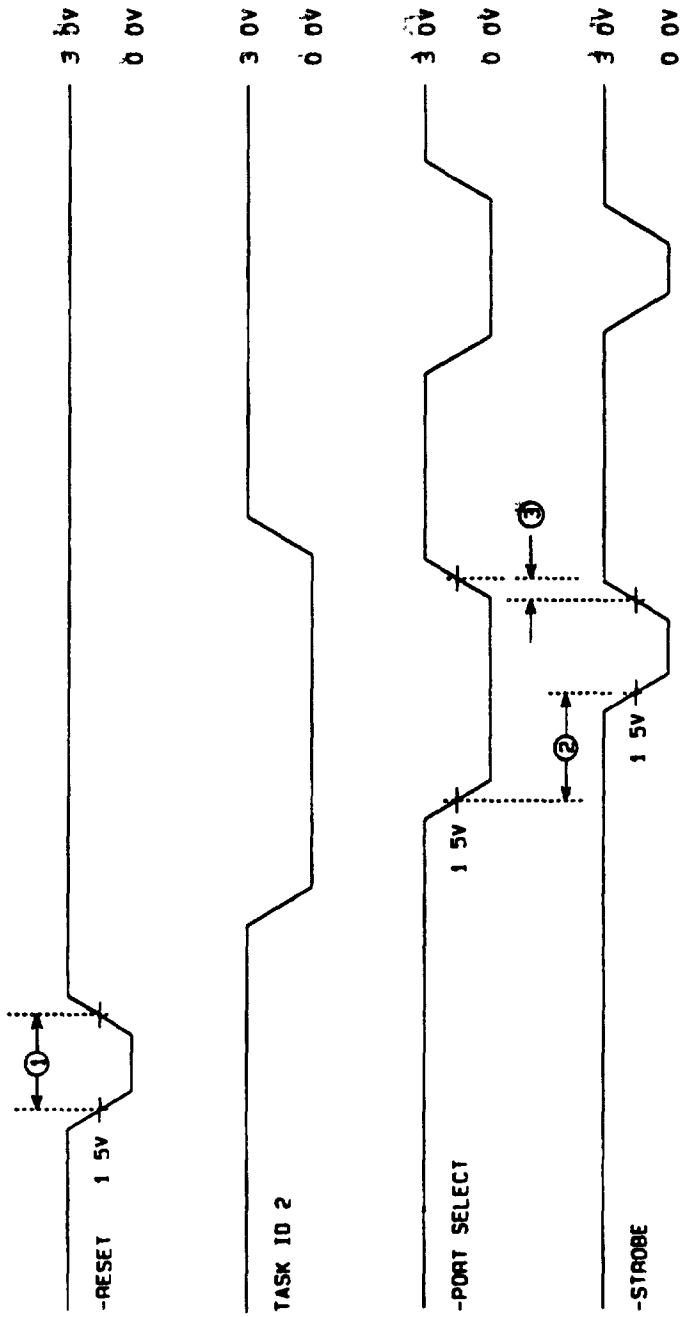


NOTE:

1. The load for each output is as specified on figure 7.

FIGURE 3. DGP operating temperature and life test circuit.

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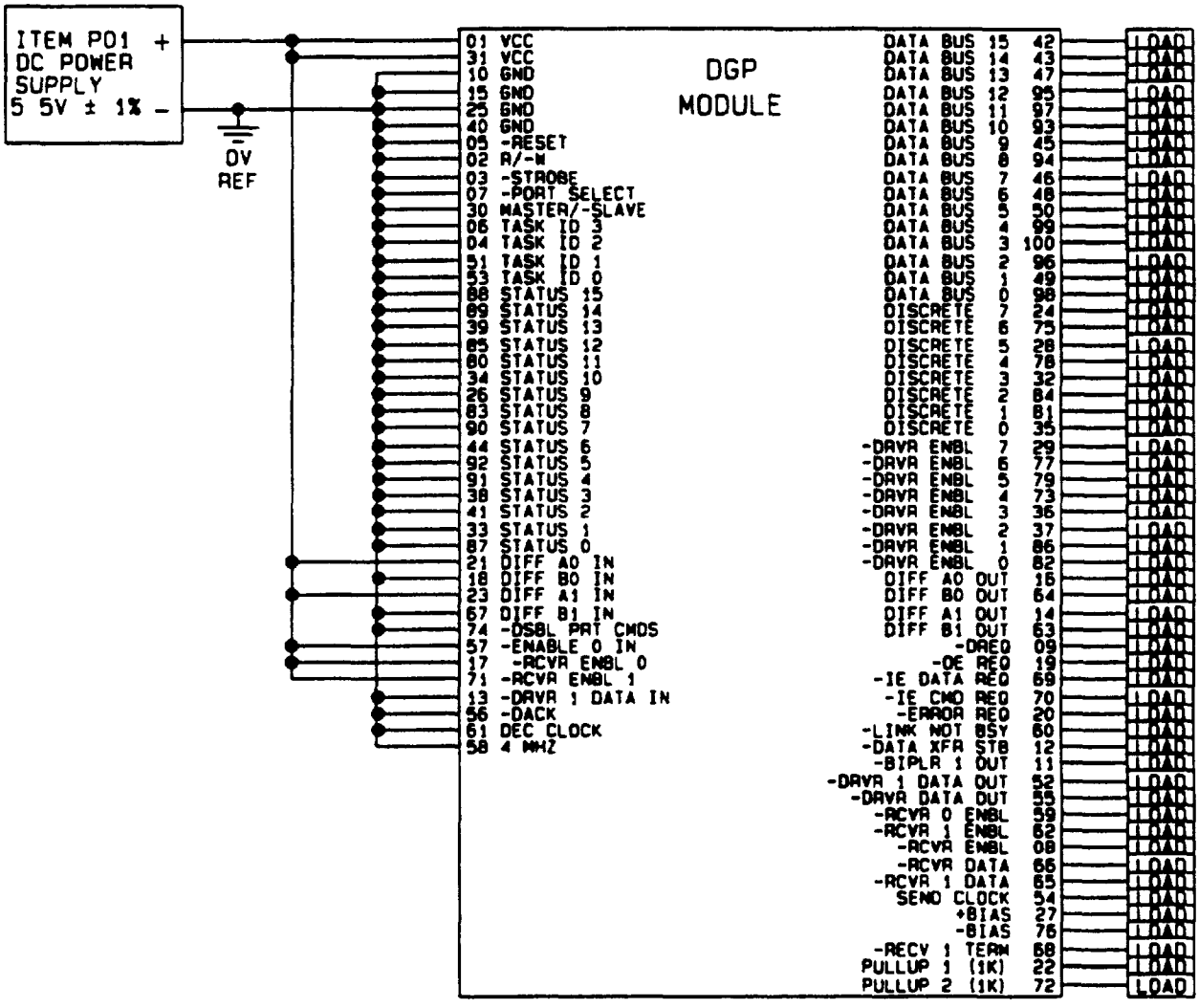


NOTES:

1. 90 nS minimum.
2. 0 nS minimum.
3. 20 nS maximum.

FIGURE 4. Operating temperature and life test waveforms.

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NOTE:

1. The load for each output is specified on figure 7.

FIGURE 5. DGP component temperature test circuit.

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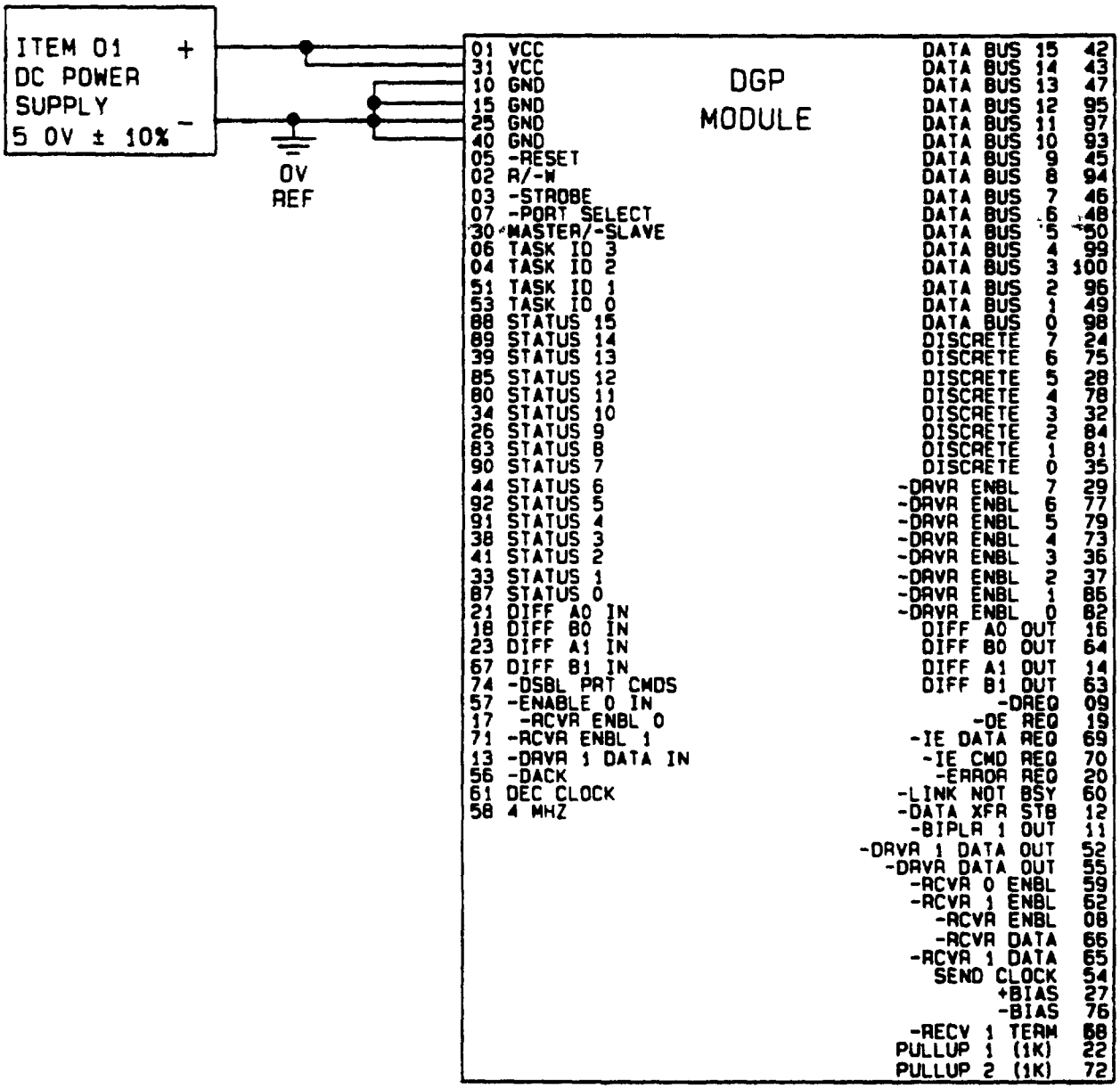


FIGURE 6. DGP vibration test circuit.

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* 4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figures 3 and 4. Following the life test and while still at the maximum class temperature, electrical tests B, G through V, Y, and Z shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements in table I. Following the return of the module to 25 ± 5°C, electrical tests B, G through V, Y and Z shall be performed in the order specified in 4.4.2 using the 25 ± 5°C ambient end-of-life requirements in table I.

4.5.4 Thermal shock test. The thermal shock test shall be conducted in accordance with MIL-M-28787. The module shall be unpowered during this test. Following the thermal shock test, the electrical tests in table I which are marked with an "X" in the test type column, shall be performed using the initial limits at 25 ± 5°C ambient.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Definitions. Negative (-) current is current directed out of the module pin and positive (+) current is current directed into the module pin. Abbreviations used in this specification are defined as follows:

(a) Cin = input capacitance.
(b) Cout = output capacitance.
(c) Ibd = input breakdown current.
(d) Icc = power supply current.
(e) Iih = input high level current.
(f) Iil = input low level current.
(g) Iin = input forcing current.
(h) Ioh = output high level current.
(i) Iol = output low level current.
(j) Ios = output short circuit current.
(k) Iozh = output high level leakage current.
(l) Iozl = output low level leakage current.
(m) NA = not applicable.
(n) PSFC = power supply filter capacitance.
(o) PSTA = power supply transient amplitude.
(p) Vic = input clamp diode voltage.
(q) Vih = input high level voltage.
(r) Vil = input low level voltage.
(s) Vin = input forcing voltage.
(t) Voh = output high level voltage.
(u) Vol = output low level voltage.
(v) Vout = output forcing voltage.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements in table I, the module is considered to have completed its useful life (see 3.9).

6.3 Module description. The following electrical parameters reflect the 0°C to 60°C end-of-life parameters in table I. Operating power supply voltage range: 4.5 V dc to 5.5 V dc. Module power dissipation: 4.068 W (max); 2.583 W (typical).

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6.3.1 Input parameters.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (μA)	Ibd max (μA)	Cin max (pF)
STATUS 0	-1.5	0.7	2.0	-0.2	20	100	25
STATUS 1							
STATUS 2							
STATUS 3							
STATUS 4							
STATUS 5							
STATUS 6							
STATUS 7							
STATUS 8							
STATUS 9							
STATUS 10							
STATUS 11							
STATUS 12							
STATUS 13							
STATUS 14							
STATUS 15							
-RESET							
TASK ID 0							
TASK ID 1							
TASK ID 2							
TASK ID 3	-1.5	0.7	2.0	-0.2	20	100	25

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (μA)	Ibd max (μA)	Cin max (pF)
-STROBE	-1.5	0.7	2.0	-0.2	20	100	25
R/-W							
-DACK							
-PORT SELECT							
DEC CLOCK							
4 MHZ	-1.5	0.7		-0.2	20	100	25
DIFF A0 IN	NA	-2.0		-1.99	NA	NA	45000
DIFF B0 IN				-0.78			45000
DIFF A1 IN				-0.9			25
DIFF B1 IN	NA	-2.0		-0.9	NA	NA	
-RCVR ENBL 0	-1.5	0.7		-0.4	20	100	
-RCVR ENBL 1		0.7		-0.4	20		
-DRVR 1 DATA IN		0.8		-1.6	40		
-ENABLE 0 IN		0.7	2.0	-0.2	20	100	
MASTER/-SLAVE		0.8	2.2	-0.01	10	10	
-DSBL PRT CMDS		0.8	2.2	-0.01	10	10	
DATA BUS 0		0.7	2.0	-0.2	20	100	
DATA BUS 1							
DATA BUS 2							
DATA BUS 3							
DATA BUS 4	-1.5	0.7	2.0	-0.2	20	100	25

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (μA)	Ibd max (μA)	Cin max (pF)
DATA BUS 5	-1.5	0.7	2.0	-0.2	20	100	25
DATA BUS 6							
DATA BUS 7							
DATA BUS 8							
DATA BUS 9							
DATA BUS 10							
DATA BUS 11							
DATA BUS 12							
DATA BUS 13							
DATA BUS 14							
DATA BUS 15	-1.5	0.7	2.0	-0.2	20	100	25

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6.3.2 Output parameters.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Iozh max (μA)	Iozl min (μA)	Ios max (mA)	Cout max (pF)
DATA BUS 0	2.4	0.4	-3.0	12.0	20	-200	-40	25
DATA BUS 1								
DATA BUS 2								
DATA BUS 3								
DATA BUS 4								
DATA BUS 5								
DATA BUS 6								
DATA BUS 7								
DATA BUS 8								
DATA BUS 9								
DATA BUS 10								
DATA BUS 11								
DATA BUS 12								
DATA BUS 13								
DATA BUS 14								
DATA BUS 15					20	-200		25
DISCRETE 0					NA	NA		NA
DISCRETE 1								
DISCRETE 2								
DISCRETE 3								
DISCRETE 4	2.4	0.4	-3.0	12.0	NA	NA	-40	NA

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6.3.2 Output parameters - Continued.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Iozh max (μA)	Iozl min (μA)	Ios max (mA)	Cout max (pF)
DISCRETE 5	2.4	0.4	-3.0	12.0	NA	NA	-40	NA
DISCRETE 6								
DISCRETE 7								
-DRVR ENBL 0								
-DRVR ENBL 1								
-DRVR ENBL 2								
-DRVR ENBL 3								
-DRVR ENBL 4								
-DRVR ENBL 5								
-DRVR ENBL 6								
-DRVR ENBL 7	2.4		-3.0	12.0	NA		-40	NA
-IE CMD REQ	NA		NA	4.0	100		NA	25
-IE DATA REQ								
-OE REQ								
-ERROR REQ	NA		NA	4.0	100		NA	25
DIFF A0 OUT	2.4		-10.0	40.0	NA		-40	NA
DIFF B0 OUT								
DIFF A1 OUT								
DIFF B1 OUT			-10.0	40.0				
-RCVR 0 ENBL			-3.0	12.0				
-RCVR 1 ENBL	2.4	0.4	-3.0	12.0	NA	NA	-40	NA

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* 6.3.2 Output parameters - Continued.

	Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Iozh max (µA)	Iozl min (µA)	Ios max (mA)	Cout max (pF)
	-RCVR ENBL	2.4	0.4	-3.0	12.0	NA	NA	-40	NA
	-LINK NOT BSY								
	-DREQ								
	-DATA XFR STB								
	SEND CLOCK								
	-DRVR 1 DATA OUT								
	-BIPLR 1 OUT			-3.0	12.0				
	-DRVR DATA OUT	2.4		-2.96	10.4	NA		-40	NA
	-RCVR 1 DATA	NA		NA	16.0	200		NA	25
	-RCVR DATA	2.4	0.4	-2.08	11.7	NA		-5.2	50
*	+BIAS	2.4	NA	-0.38	NA			NA	NA
	-BIAS	NA	0.4	NA	0.1				
	-RECV 1 TERM	NA	NA	NA	NA				
*	PULLUP 1	2.4		-1.9					
*	PULLUP 2	2.4	NA	-1.9	NA	NA	NA	NA	NA

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6.3.3 Dynamic parameters. The dynamic parameters appear with the dynamic test.

6.4 Module function. The DGP module contains a semi-custom Input/Output (I/O) port gate array to provide bidirectional data transfers between interconnected microprocessor based subsystems. The two subsystems' interfaces to the DGP module are through the intersubsystem and intrasubsystem links. The intersubsystem interface (serial side) consists of dc differential line drivers and receivers for communication up to 250 feet of twisted shielded pair cable. Serial interface protocol is Manchester encoded, 20 bit command/data words at a 1 MHZ rate. Up to two serial side interfaces can be accommodated by a single DGP module with expansion of up to eight interfaces possible with the addition of a line driver/receiver module. Selection of the driver receiver pairs on the added module can be controlled by enable signals from the DGP module. The intrasubsystem interface (parallel side) consists of both command data and control links with the subsystem. Command/data links consist of sixteen bidirectional interfaces to the subsystem data bus, called the Bus Interface Unit (BIU), sixteen status and discrete inputs for general purpose use, called the Status and Discrete Interface (SDI), and eight discrete outputs for general purpose use. Control links consist of:

(a) Four task IDs which specify the port command from the subsystem.

(b) Three control signals, -STROBE, -PORT SELECT, and R/-W, which are decoded in conjunction with the task IDs.

(c) Two Direct Memory Access (DMA) control signals, -DREQ and -DACK, used for high speed data transfers.

(d) Four port interrupts, -IE CMD REQ, -IE DATA REQ, -OE REQ, and -ERROR REQ, to identify command data conditions on the port.

(e) Two port status functions, -LINK NOT BSY and -DATA XFR STB, to identify certain port operations.

(f) One control input, -RESET, for implementing a port hardware reset.

The DGP module provides for various data transfers between the serial side and parallel side interfaces, as well as between certain registers within the I/O port. The port status register monitors the present status of the I/O port and can be read from

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the serial or parallel side. The Test and Interrupt Control register (TIC) sets up the port for DMA operations and configures certain test modes. The TIC register can be read from or written to from the parallel side interface, but can only be read from the serial side interface. The intersubsystem link register (ISL) controls the line driver and receiver enables for the serial side interface, and can be read from the parallel side. The input element (IE) is the pair of registers in the DGP module used to receive serial data words from the ISL. The input element data can be transferred to the BIU through a parallel side command or DMA operation. The output element (OE) is the pair of registers in the DGP module used to hold serial data words prior to transmission over the ISL. The DGP module also contains several built in diagnostic features for subsystem evaluation. By sending the appropriate serial side command or by writing the appropriate bits in the TIC register, the module can be set up in the serial side loop test mode or the parallel side loop test mode. In the serial side loop test mode, all data received by the DGP module over its line receivers is echoed back over its line drivers. This feature is used to evaluate the DGP module and the serial data link from the serial side. In the parallel side loop test mode, all data received by the DGP module from the parallel side interface is echoed back over the BIU. This feature is used to evaluate the DGP module from the parallel side.

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6.4.1 Functional description of I/O pins.

Signal name	Pin number	I/O	Functional description
STATUS 0	87	I	16-bit general purpose input bus which can be read by a parallel side or serial side command.
STATUS 1	33		
STATUS 2	41		
STATUS 3	38		
STATUS 4	91		
STATUS 5	92		
STATUS 6	44		
STATUS 7	90		
STATUS 8	83		
STATUS 9	26		
STATUS 10	34		
STATUS 11	80		
STATUS 12	85		
STATUS 13	39		
STATUS 14	89		
STATUS 15	88		
-RESET	5		Module input used to initialize the DGP module.
TASK ID 0	53		4-bit code which selects the parallel side command that the I/O port will perform.
TASK ID 1	51		
TASK ID 2	4		
TASK ID 3	6		
-STROBE	3		Module input used to enable data to the BIU during a read. During a write, this input is used to enter data to the DGP module.
R/-W	2		Module input to define a parallel side command as a read or write.
-DACK	56	I	DMA input used to indicate that the DMA is to be performed.

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6.4.1 Functional description of I/O pins - Continued.

Signal name	Pin number	I/O	Functional description
-PORT SELECT	7	I	Module input used to enable parallel side commands to the DGP module.
DEC CLOCK	61		12 MHz clock input used by Manchester encoder/decoder to read incoming data streams.
4 MHZ	58		4 MHz clock input used for internal timing operations on the DGP module.
DIFF A0 IN DIFF B0 IN	21 18		Differential module input to line receiver 0.
DIFF A1 IN DIFF B1 IN	23 67		Differential module input to line receiver 1.
-RCVR ENBL 0	17		-Receiver Enable 0 - Module input used to enable line receiver 0.
-RCVR ENBL 1	71		-Receiver Enable 1 - Module input used to enable line receiver 1.
-DRVR 1 DATA IN	13		-Driver 1 Data In - Module data input to line driver.
-ENABLE 0 IN	57		Module input used to enable line driver 0.
MASTER/-SLAVE	30		Module input used to program the DGP module as a master or a slave.
-DSBL PRT CMDS	74	I	-Disable Port CMDS - Module input used to force DGP module to not execute port commands.

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6.4.1 Functional description of I/O pins - Continued.

Signal name	Pin number	I/O	Functional description
DATA BUS 0	98	I/O	16 bit, bidirectional, 3-state bus which comprises the parallel data bus to the DGP module.
DATA BUS 1	49		
DATA BUS 2	96		
DATA BUS 3	100		
DATA BUS 4	99		
DATA BUS 5	50		
DATA BUS 6	48		
DATA BUS 7	46		
DATA BUS 8	94		
DATA BUS 9	45		
DATA BUS 10	93		
DATA BUS 11	97		
DATA BUS 12	95		
DATA BUS 13	47		
DATA BUS 14	43		
DATA BUS 15	42	I/O	
DISCRETE 0	35	O	8-bit general purpose output register which can be written by a serial side command.
DISCRETE 1	81		
DISCRETE 2	84		
DISCRETE 3	32		
DISCRETE 4	78		
DISCRETE 5	28		
DISCRETE 6	75		
DISCRETE 7	24		
-DRVR ENBL 0	82		-Driver Enable 0-7 - Module outputs used to enable line drivers 0 through 7. The driver enables can be set through a parallel side command.
-DRVR ENBL 1	86		
-DRVR ENBL 2	37		
-DRVR ENBL 3	36		
-DRVR ENBL 4	73		
-DRVR ENBL 5	79		
-DRVR ENBL 6	77		
-DRVR ENBL 7	29		
-IE CMD REQ	70	O	Interrupt output which, when enabled, will go active when the IEHR contains a command word.

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6.4.1 Functional description of I/O pins - Continued.

Signal name	Pin number	I/O	Functional description
-IE DATA REQ	69	O	Interrupt output which, when enabled, will go active when the IEHR contains a data word.
-OE REQ	19		Interrupt output which, when enabled, will go active when the OEHR is empty.
-ERROR REQ	20		Interrupt output which, when enabled, will go active when the OE is overwritten, transmission error occurs, or a data bus conflict occurs.
DIFF A0 OUT DIFF B0 OUT	16 64		Differential module output from line driver 0.
DIFF A1 OUT DIFF B1 OUT	14 63		Differential module output from line driver 1.
-RCVR 0 ENBL	59		-Receiver 0 Enable - Module output which will go low when both -RCVR ENBL and -DRVR ENBL 0 go low.
-RCVR 1 ENBL	62		-Receiver 0 Enable - Module output which will go low when both -RCVR ENBL and -DRVR ENBL 1 go low.
-RCVR ENBL	8		-Receiver Enable - Module output used to enable up to 8 line receivers in conjunction with the driver enables.
-LINK NOT BSY	60	O	-Link Not Busy - Module output which will go active when the ISL is ready to receive a transmission.

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6.4.1 Functional description of I/O pins - Continued.

Signal name	Pin number	I/O	Functional description
-DREQ	9	O	DMA output which will go active when the DGP module is enabled for a DMA transfer.
-DATA XFR STB	12		-Data Transfer Strobe - Module output which will go active when the DGP module is in the BIU output mode and data is available in the IEHR, or when the module has received a BIU serial command.
SEND CLOCK	54		2 MHz clock output of Manchester encoder/decoder used to shift out serial data stream.
-DRVR 1 DATA OUT	52		-Drive 1 Data Out - Manchester encoded module data output of the DGP module intended for line driver 1.
-BIPLR 1 OUT	11		-Bipolar 1 Out - Manchester encoded module data output transmitted by the DGP module.
-DRVR DATA OUT	55		-Driver Data Out - Manchester encoded module data output of the DGP modules.
-RCVR 1 DATA	65		-Receiver 1 Data - Module data output of line receiver 1.
-RCVR DATA	66		-Receiver Data - Module output of line receiver 0.
+BIAS	27		Module output used to dc bias line receiver 1 consisting of a 4.7 KOhm resistor to Vcc.
-BIAS	76	O	Module output used to dc bias line receiver 1 consisting of a 3.3 KOhm resistor to GND.

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6.4.1 Functional description of I/O pins - Continued.

Signal name	Pin number	I/O	Functional description
PULLUP 1 (1K) PULLUP 2 (1K)	22 72	O	1 KOhm resistor to Vcc. Can be used to pull-up unused module pins.
-RECV 1 TERM	68	O	This output consists of a 75 Ohm resistor in series with a 22.000 pF to DIFF A1 IN. It can be used to ac terminate differential line receiver 1 by connecting it to DIFF B1 IN.

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6.5 Description of tests. The following tables and associated figures describe the tests to be performed.

(a) Isolation test - table II.

(b) Continuity test - table III.

(c) Vcc stress test - table IV.

(d) Ibd test - table V.

(e) Vic test - table VI.

(f) Ios test - table VII.

(g) Iih test - table VIII.

(h) Iil test - table IX.

(i) Voh test - table X, figure 7.

(j) Vol test - table XI, figure 7.

(k) Iozh test - table XII.

(l) Iozl test - table XIII.

(m) Icc test - table XIV.

(n) Functional test - table XV, figure 8.

(o) Dynamic test - tables XVI and XVII, figures 9 and 10.

(p) PSFC test - table XVIII.

(q) PSTA test - table XIX, figures 8 and 11.

(r) Output crosstalk tests - tables XX and XXI, figures 8, 12 and 13.

(s) Cin test - table XXII.

(t) Cout test - table XXIII.

(u) Threshold test - table XXIV, figure 8.

(v) Disable time test - table XXV, figures 8, 14 and 15.

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6.6 Functional test vectors. Functional test vectors are not printed herein because of their extreme length; and, even when presented as a hard copy, they require conversion to on-line format. When the entire sequence of vector groups is needed for driving devices as part of inspection, a test program output should be generated in a form compatible with test system architecture (magnetic tape format usually preferred). For reference purposes, the Naval Surface Warfare Center, Crane Division (Code 603) maintains a limited quantity of dated, hard copy printouts of the functional test vectors and these may be obtained upon request.

* 6.7 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

*Custodians:
Army - ER
Navy - NW
Air Force - 85

Review activities:
Army - AT
Navy - AS, MC
Air Force - 13, 19
DLA - ES

Preparing activity:
Navy - NW

(Project 5963-0296)

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TABLE II. Isolation test.

Test cond	Test description	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
A	Isolation	All pins	Frame	10	INF	MOhm	1-7

NOTES:

1. Isolation test tolerance table.

Condition	Value	Accuracy	Test condition
Megohm bridge voltage	100 V	± 10 percent	A
Isolation measurement (see note 6)	NA NA	± 500 KOhm ± 50 KOhm	A A

2. The ohmmeter used shall be item R06.

3. In the setup conditions in table II, 'A' shall correspond to the positive terminal of the megohm bridge, and 'B' shall correspond to the negative terminal.

4. Warning: When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "DISCHARGE" before connecting or disconnecting the test leads.

5. When testing for frame isolation, penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal of the megohm bridge.

6. The minimum end-of-life limit is 1 MOhm.

7. Abbreviations used are defined in 6.1.

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TABLE III. Continuity test.

Test cond	Test description	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
B1	Vcc	1	31	0.0	1.0	Ohm	1-4
B2	GND	10	15				
B3	GND	15	25				
B4	GND	25	40		1.0		
B5	Vcc	1	31		0.1		
B6	GND	10	15				
B7	GND	15	25				
B8	GND	25	40	0.0	0.1	Ohm	1-4

NOTES:

1. Continuity test tolerance table.

Condition	Value	Accuracy	Test condition
Continuity measurement	NA NA	± 50 MOhm ± 5 MOhm	B1-B4 B5-B8

2. The ohmmeter used shall be item R07.

3. In the setup conditions in table III, 'A' shall correspond to the positive terminal of the ohmmeter, and 'B' shall correspond to the negative terminal.

4. Abbreviations used are defined in 6.1.

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* TABLE IV. Power supply stress test.

Test cond	Power supply	Vin (V)	Connect Positive lead to	Connect Negative lead to	25 ± 5°C Initial limits	Notes
* C	Vcc	7.0	1,31	10,15 25,40	Module shall pass all remaining tests.	1-6

NOTES:

1. Power supply stress test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	7.0 V	+0 mV, -70 mV	C
Power supply current limit	1.269 A	NA	C

2. There shall be no power supply voltage overshoot on this test.

3. Connect the power supplies to the module as indicated in table IV.

4. When performing the power supply stress test, no measurements are taken. This condition shall be maintained for a minimum of 5 seconds.

5. Abbreviations used are defined in 6.1.

*6. Pins 30 (MASTER/-SLAVE) and 74 (-DISABLE PORT CMDS) shall be tied to 0.0V during this test.

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TABLE V. Input breakdown current test.

Test cond	Signal name	Pin no.	Vin (V)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
D1	STATUS 0	87	5.5	0	100	µA	1-3
D2	STATUS 1	33					
D3	STATUS 2	41					
D4	STATUS 3	38					
D5	STATUS 4	91					
D6	STATUS 5	92					
D7	STATUS 6	44					
D8	STATUS 7	90					
D9	STATUS 8	83					
D10	STATUS 9	26					
D11	STATUS 10	34					
D12	STATUS 11	80					
D13	STATUS 12	85					
D14	STATUS 13	39					
D15	STATUS 14	89					
D16	STATUS 15	88					
D17	-RESET	5					
D18	TASK ID 0	53					
D19	TASK ID 1	51					
D20	TASK ID 2	4	5.5	0	100	µA	1-3

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TABLE V. Input breakdown current test - Continued.

Test cond	Signal name	Pin no.	Vin (V)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
D21	TASK ID 3	6	5.5	0	100	µA	1-3
D22	-STROBE	3					
D23	R/-W	2					
D24	-DACK	56					
D25	-PORT SELECT	7					
D26	DEC CLOCK	61					
D27	4 MHZ	58					
D28	-RCVR ENBL 0	17					
D29	-RCVR ENBL 1	71					
D30	-DRVR 1 DATA IN	13					
D31	-ENABLE 0 IN	57			100		
D32	MASTER/-SLAVE	30			10		
D33	-DSBL PRT CMDS	74			10		
D34	DATA BUS 0	98			100		
D35	DATA BUS 1	49					
D36	DATA BUS 2	96					
D37	DATA BUS 3	100					
D38	DATA BUS 4	99					
D39	DATA BUS 5	50					
D40	DATA BUS 6	48					
D41	DATA BUS 7	46	5.5	0	100	µA	1-3

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TABLE V. Input breakdown current test - Continued.

Test cond	Signal name	Pin no.	Vin (V)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
D42	DATA BUS 8	94	5.5	0	100	µA	1-3
D43	DATA BUS 9	45					
D44	DATA BUS 10	93					
D45	DATA BUS 11	97					
D46	DATA BUS 12	95					
D47	DATA BUS 13	47					
D48	DATA BUS 14	43					
D49	DATA BUS 15	42	5.5	0	100	µA	1-3

NOTES:

1. Input breakdown current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	D1-D49
Power supply current limit	1.058 A	NA	D1-D49
Forcing voltage, (Vin)	5.5 V	± 55 mV	D1-D49
Pattern voltage high, (VH)	3.0 V	± 100 mV	D1-D49
Pattern voltage low, (VL)	0.0 V	± 100 mV	D1-D49

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Input current measurement	NA | NA	± 5.0 µA ± 0.5 µA ± 5.0 µA	D1-D31 D32,D33 D34-D49

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table V and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE VI. Input clamp diode voltage test.

Test cond	Signal name	Pin no.	Iin (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
E1	STATUS 0	87	-18.0	-1.5	-0.4	V	1-3
E2	STATUS 1	33					
E3	STATUS 2	41					
E4	STATUS 3	38					
E5	STATUS 4	91					
E6	STATUS 5	92					
E7	STATUS 6	44					
E8	STATUS 7	90					
E9	STATUS 8	83					
E10	STATUS 9	26					
E11	STATUS 10	34					
E12	STATUS 11	80					
E13	STATUS 12	85					
E14	STATUS 13	39					
E15	STATUS 14	89					
E16	STATUS 15	88					
E17	-RESET	5					
E18	TASK ID 0	53					
E19	TASK ID 1	51					
E20	TASK ID 2	4	-18.0	-1.5	-0.4	V	1-3

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TABLE VI. Input clamp diode voltage test - Continued.

Test cond	Signal name	Pin no.	Iin (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
E21	TASK ID 3	6	-18.0	-1.5	-0.4	V	1-3
E22	-STROBE	3					
E23	R/-W	2					
E24	-DACK	56					
E25	-PORT SELECT	7					
E26	DEC CLOCK	61					
E27	4 MHZ	58					
E28	-RCVR ENBL 0	17					
E29	-RCVR ENBL 1	71	-18.0				
E30	-DRVR 1 DATA IN	13	-12.0				
E31	-ENABLE 0 IN	57	-18.0				
E32	DATA BUS 0	98					
E33	DATA BUS 1	49					
E34	DATA BUS 2	96					
E35	DATA BUS 3	100					
E36	DATA BUS 4	99					
E37	DATA BUS 5	50					
E38	DATA BUS 6	48					
E39	DATA BUS 7	46					
E40	DATA BUS 8	94	-18.0	-1.5	-0.4	V	1-3

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TABLE VI. Input clamp diode voltage test - Continued.

Test cond	Signal name	Pin no.	Iin (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
E41	DATA BUS 9	45	-18.0	-1.5	-0.4	V	1-3
E42	DATA BUS 10	93					
E43	DATA BUS 11	97					
E44	DATA BUS 12	95					
E45	DATA BUS 13	47					
E46	DATA BUS 14	43					
E47	DATA BUS 15	42	-18.0	-1.5	-0.4	V	1-3

NOTES:

1. Input clamp diode voltage test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V	± 45 mV	E1-E47
Power supply current limit	1.058 A	NA	E1-E47
Forcing current, (Iin)	-18 mA -12 mA	± 1.8 mA ± 1.2 mA	E1-E29, E31-E47 E30
Pattern voltage high, (VH) low, (VL)	3.0 V 0.0 V	± 100 mV ± 100 mV	E1-E47 E1-E47
Input voltage measurement	NA	± 55 mV	E1-E47

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing current shall be applied to the pin specified in the appropriate test conditions in table VI and the voltage then measured.

3. Abbreviations used are defined in 6.1.

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TABLE VII. Output short circuit current test.

Test cond	Signal name	Pin no.	Vout (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
F1	DATA BUS 0	98	0.0	-225.0	-40.0	mA	1-4
F2	DATA BUS 1	49					
F3	DATA BUS 2	96					
F4	DATA BUS 3	100					
F5	DATA BUS 4	99					
F6	DATA BUS 5	50					
F7	DATA BUS 6	48					
F8	DATA BUS 7	46					
F9	DATA BUS 8	94					
F10	DATA BUS 9	45					
F11	DATA BUS 10	93					
F12	DATA BUS 11	97					
F13	DATA BUS 12	95					
F14	DATA BUS 13	47					
F15	DATA BUS 14	43					
F16	DATA BUS 15	42					
F17	DISCRETE 0	35					
F18	DISCRETE 1	81					
F19	DISCRETE 2	84					
F20	DISCRETE 3	32	0.0	-225.0	-40.0	mA	1-4

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TABLE VII. Output short circuit current test - Continued.

Test cond	Signal name	Pin no.	Vout (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
F21	DISCRETE 4	78	0.0	-225.0	-40.0	mA	1-4
F22	DISCRETE 5	28					
F23	DISCRETE 6	75					
F24	DISCRETE 7	24					
F25	-DRVR ENBL 0	82					
F26	-DRVR ENBL 1	86					
F27	-DRVR ENBL 2	37					
F28	-DRVR ENBL 3	36					
F29	-DRVR ENBL 4	73					
F30	-DRVR ENBL 5	79					
F31	-DRVR ENBL 6	77					
F32	-DRVR ENBL 7	29		-225.0			
F33	DIFF A0 OUT	16		-120.0			
F34	DIFF B0 OUT	64					
F35	DIFF A1 OUT	14					
F36	DIFF B1 OUT	63		-120.0			
F37	-RCVR 0 ENBL	59		-225.0			
F38	-RCVR 1 ENBL	62					
F39	-RCVR ENBL	8					
F40	-LINK NOT BSY	60	0.0	-225.0	-40.0	mA	1-4

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TABLE VII. Output short circuit current test - Continued.

Test cond	Signal name	Pin no.	Vout (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
F41	-DREQ	9	0.0	-225.0	-40.0	mA	1-4
F42	-DATA XFR STB	12					
F43	SEND CLOCK	54					
F44	-DRVR 1 DATA OUT	52					
F45	-BIPLR 1 OUT	11					
F46	-DRVR DATA OUT	55		-225.0	-40.0		
F47	-RCVR DATA	66	0.0	-6.3	-5.2	mA	1-4

NOTES:

1. Output short circuit current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	F1-F47
Power supply current limit	1.058 A	NA	F1-F47
Forcing voltage, (Vout)	0.0 V	± 20 mV	F1-F47
Pattern voltage high, (VH)	3.0 V	± 100 mV	F1-F47
low, (VL)	0.0 V	± 100 mV	F1-F47

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Output current measurement	NA	± 9.25 mA	F1-F32
		± 4.0 mA	F33-F36
		± 9.25 mA	F37-F46
	NA	± 55.0 µA	F47

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VII and the current then measured. The forcing voltage undershoot shall not exceed 100 mV.

3. When performing this test, only one output shall be tested at a time, and the duration of momentarily grounding each output shall not exceed 1 second.

4. Abbreviations used are defined in 6.1.

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TABLE VIII. Input current, high level, test.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
G1	STATUS 0	87	2.7	0	20	µA	1-3
G2	STATUS 1	33					
G3	STATUS 2	41					
G4	STATUS 3	38					
G5	STATUS 4	91					
G6	STATUS 5	92					
G7	STATUS 6	44					
G8	STATUS 7	90					
G9	STATUS 8	83					
G10	STATUS 9	26					
G11	STATUS 10	34					
G12	STATUS 11	80					
G13	STATUS 12	85					
G14	STATUS 13	39					
G15	STATUS 14	89					
G16	STATUS 15	88					
G17	-RESET	5					
G18	TASK ID 0	53					
G19	TASK ID 1	51					
G20	TASK ID 2	4	2.7	0	20	µA	1-3

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TABLE VIII. Input current, high level, test - Continued.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
G21	TASK ID 3	6	2.7	0	20	µA	1-3
G22	-STROBE	3					
G23	R/-W	2					
G24	-DACK	56					
G25	-PORT SELECT	7					
G26	DEC CLOCK	61					
G27	4 MHZ	58					
G28	-RCVR ENBL 0	17					
G29	-RCVR ENBL 1	71			20		
G30	-DRVR 1 DATA IN	13			40		
G31	-ENABLE 0 IN	57		0	20		
G32	MASTER/-SLAVE	30		-10	10		
G33	-DSBL PRT CMDS	74		-10	10		
G34	DATA BUS 0	98		0	20		
G35	DATA BUS 1	49					
G36	DATA BUS 2	96					
G37	DATA BUS 3	100					
G38	DATA BUS 4	99					
G39	DATA BUS 5	50					
G40	DATA BUS 6	48	2.7	0	20	µA	1-3

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TABLE VIII. Input current, high level, test - Continued.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
G41	DATA BUS 7	46	2.7	0	20	µA	1-3
G42	DATA BUS 8	94					
G43	DATA BUS 9	45					
G44	DATA BUS 10	93					
G45	DATA BUS 11	97					
G46	DATA BUS 12	95					
G47	DATA BUS 13	47					
G48	DATA BUS 14	43					
G49	DATA BUS 15	42	2.7	0	20	µA	1-3

NOTES:

1. Input current, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	G1-G49
Power supply current limit	1.058 A	NA	G1-G49
Forcing voltage, (Vih)	2.4 V	± 120 mV	G1-G49
Pattern voltage high, (VH)	3.0 V	± 100 mV	G1-G49
low, (VL)	0.0 V	± 100 mV	G1-G49

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Input current measurement	NA	± 1 μA	G1-G29
		± 2 μA	G30
		± 1 μA	G31
		± 1 μA	G32,G33
	NA	± 1 μA	G34-G49

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VIII and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE IX. Input current, low level, test.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
H1	STATUS 0	87	0.4	-200	-5	µA	1-3
H2	STATUS 1	33					
H3	STATUS 2	41					
H4	STATUS 3	38					
H5	STATUS 4	91					
H6	STATUS 5	92					
H7	STATUS 6	44					
H8	STATUS 7	90					
H9	STATUS 8	83					
H10	STATUS 9	26					
H11	STATUS 10	34					
H12	STATUS 11	80					
H13	STATUS 12	85					
H14	STATUS 13	39					
H15	STATUS 14	89					
H16	STATUS 15	88					
H17	-RESET	5					
H18	TASK ID 0	53					
H19	TASK ID 1	51					
H20	TASK ID 2	4	0.4	-200	-5	µA	1-3

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TABLE IX. Input current, low level, test - Continued.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
H21	TASK ID 3	6	0.4	-200	-5	µA	1-3
H22	-STROBE	3					
H23	R/-W	2					
H24	-DACK	56					
H25	-PORT SELECT	7					
H26	DEC CLOCK	61					
H27	4 MHZ	58		-200	-5		
H28	DIFF A0 IN	21		-1985	-1285		
H29	DIFF B0 IN	18		-780	-80		
H30	DIFF A1 IN	23		-900	-200		
H31	DIFF B1 IN	67		-900	-200		
H32	-RCVR ENBL 0	17		-400	-30		
H33	-RCVR ENBL 1	71		-400	-30		
H34	-DRVR 1 DATA IN	13		-1600	-700		
H35	-ENABLE 0 IN	57		-200	-5		
H36	MASTER/-SLAVE	30		-10	10		
H37	-DSBL PRT CMDS	74		-10	10		
H38	DATA BUS 0	98		-200	-5		
H39	DATA BUS 1	49					
H40	DATA BUS 2	96	0.4	-200	-5	µA	1-3

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TABLE IX. Input current, low level, test - Continued.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
H41	DATA BUS 3	100	0.4	-200	-5	µA	1-3
H42	DATA BUS 4	99					
H43	DATA BUS 5	50					
H44	DATA BUS 6	48					
H45	DATA BUS 7	46					
H46	DATA BUS 8	94					
H47	DATA BUS 9	45					
H48	DATA BUS 10	93					
H49	DATA BUS 11	97					
H50	DATA BUS 12	95					
H51	DATA BUS 13	47					
H52	DATA BUS 14	43					
H53	DATA BUS 15	42	0.4	-200	-5	µA	1-3

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NOTES:

1. Input current, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	H1-H53
Power supply current limit	1.058 A	NA	H1-H53
Forcing voltage, (Vil)	0.4 V	± 20 mV	H1-H53
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 H1-H53 H1-H53
Input current measurement	NA NA	± 9.75 µA ± 35.0 µA ± 35.0 µA ± 35.0 µA ± 18.5 µA ± 45.0 µA ± 9.75 µA ± 1.0 µA ± 9.75 µA	H1-H27 H28 H29 H30,H31 H32,H33 H34 H35 H36,H37 H38-H53

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

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NOTES: - Continued.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table IX and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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* TABLE X. Output voltage, high level, test.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
J1	DATA BUS 0	98	-3.0	2.4	4.5	V	1-5
J2	DATA BUS 1	49					
J3	DATA BUS 2	96					
J4	DATA BUS 3	100					
J5	DATA BUS 4	99					
J6	DATA BUS 5	50					
J7	DATA BUS 6	48					
J8	DATA BUS 7	46					
J9	DATA BUS 8	94					
J10	DATA BUS 9	45					
J11	DATA BUS 10	93					
J12	DATA BUS 11	97					
J13	DATA BUS 12	95					
J14	DATA BUS 13	47					
J15	DATA BUS 14	43					
J16	DATA BUS 15	42					
J17	DISCRETE 0	35					
J18	DISCRETE 1	81					
J19	DISCRETE 2	84					
J20	DISCRETE 3	32	-3.0	2.4	4.5	V	1-5

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* TABLE X. Output voltage, high level, test - Continued.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
J21	DISCRETE 4	78	-3.0	2.4	4.5	V	1-5
J22	DISCRETE 5	28					
J23	DISCRETE 6	75					
J24	DISCRETE 7	24					
J25	-DRVR ENBL 0	82					
J26	-DRVR ENBL 1	86					
J27	-DRVR ENBL 2	37					
J28	-DRVR ENBL 3	36					
J29	-DRVR ENBL 4	73					
J30	-DRVR ENBL 5	79					
J31	-DRVR ENBL 6	77					
J32	-DRVR ENBL 7	29	-3.0				
J33	DIFF A0 OUT	16	-10.0				
J34	DIFF B0 OUT	64					
J35	DIFF A1 OUT	14					
J36	DIFF B1 OUT	63	-10.0				
J37	-RCVR 0 ENBL	59	-3.0				
J38	-RCVR 1 ENBL	62					
J39	-RCVR ENBL	8					
J40	-LINK NOT BSY	60	-3.0	2.4	4.5	V	1-5

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* TABLE X. Output voltage, high level, test - Continued.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
J41	-DREQ	9	-3.0	2.4	4.5	V	1-5
J42	-DATA XFR STB	12					
J43	SEND CLOCK	54					
J44	-DRVR 1 DATA OUT	52					
J45	-BIPLR 1 OUT	11	-3.0				
J46	-DRVR DATA OUT	55	-2.96				
J47	-RCVR DATA	66	-1.68		4.5		
J48	+BIAS	27	-0.38		3.0		
J49	PULLUP 1 (1K)	22	-1.9		2.8		
J50	PULLUP 2 (1K)	72	-1.9	2.4	2.8	V	1-5

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NOTES:

1. Output voltage, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V	± 45 mV	J1-J50
Power supply current limit	1.058 A	NA	J1-J50
Forcing current, (Ioh)	-3.0 µA -10.0 µA -3.0 µA -2.96 µA -1.68 µA -0.38 µA -1.9 µA	± 30.0 µA ± 100.0 µA ± 30.0 µA ± 29.6 µA ± 16.8 µA ± 4.0 µA ± 19.0 µA	J1-J32 J33-J36 J37-J45 J46 J47 J48 J49,J50
* Pattern voltage high, (VH) low, (VL)	 2.0,2.2 V 0.8,0.7 V	 ± 20 mV,± 22 mV ± 8 mV,± 7 mV	 J1-J50 J1-J50
* Output voltage measurement	NA | NA	± 105 mV ± 30 mV ± 20 mV	J1-J47 J48 J49,J50

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

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NOTES: - Continued.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

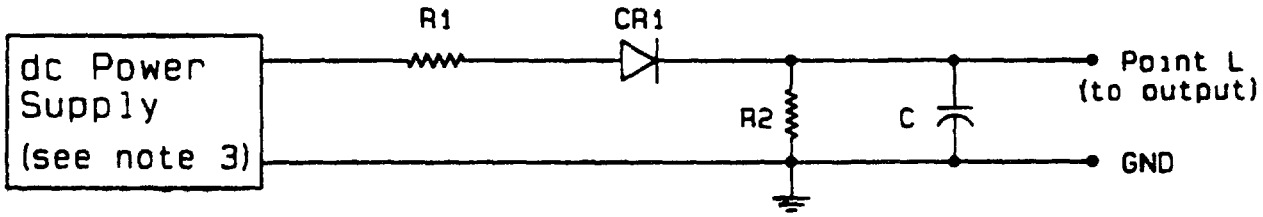
(d) The forcing current shall be applied to the pin specified in the appropriate test conditions in table X and the voltage then measured.

3. The load shall be as indicated on figure 7.

4. Abbreviations used are defined in 6.1.

*5. For pattern voltage, the high and low level input threshold voltage for each pin is as specified in 6.3.1.

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NOTES:

1. CR1 = 1N4148 or equivalent.

2. The load capacitance is defined as follows:

    (a) C = 150 ± 15 pF (including probe and parasitic capacitance) for sample dynamic tests.

    (b) C = 50 ± 5 pF (including probe and parasitic capacitance) for crosstalk and disable time tests.

    (c) C = 300 ± 30 pF (including probe and parasitic capacitance) for qualification dynamic tests.

    (d) The functional test shall be performed with only a 150 ± 15 pF load (including probe and parasitic capacitance) attached to each output.

    (e) C not required for any other tests.

3. The above load is for all output pins, except 19, 20, 65, 69, 70, 76, and 68.

4. The load power supply voltage shall be 2.1 V ± 1 percent for the output disable time test and 2.5 V ± 1 percent for all other tests.

FIGURE 7a. Logic load circuit.

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NOTES: - Continued.

5. Select R1 for a current flow of Iol when point L is held at Vol with the load power supply set at 2.5 V ± 1 percent. The values of Iol and Vol for each output are listed below.

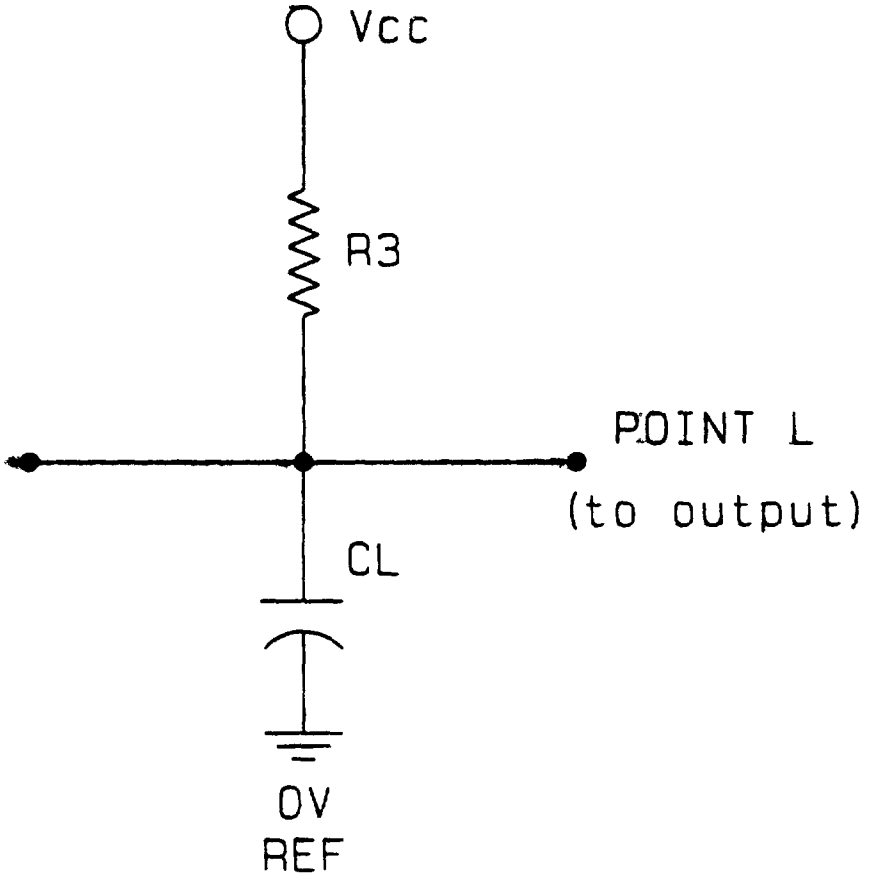
6. The value of R2 for each output load shall be as specified below.

Pin number	R2 1/ (ohms)	Iol (mA)	Vol (volts)
8,9,11,12,24,28,29,32,35,36,37, 42,43,45,46,47,48,49,50,52,54, 59,60,62,73,75,77,78,79,81,82, 84,86,93,94,95,96,97,98,99,100	800	12.0	0.4
14,16,63,64	240	40.0	0.4
55	811	10.4	0.4
66	1154	11.7	0.4
22,72	1200	NA	NA
27	6316	NA	NA

1/ All resistors are metal film, ± 1 percent, 1/4 W.

FIGURE 7a. Logic load circuit - Continued.

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NOTES:

1. This load is required for output pins 19, 20, 65, 69, 70, and 76. (Open collector outputs and -BIAS.)

   (a) The value of R3 for pins 19, 20, 69, and 70 is 1275 ohms.

   (b) The value of R3 for pin 65 is 319 ohms.

   (c) The value of R3 for pin 76 is 51000 ohms.

2. The load capacitance is defined as follows.

   (a) C = 150 ± 15 pF (including probe and parasitic capacitance) for sample dynamic and functional tests.

   (b) C = 50 ± 5 pF (including probe and parasitic capacitance) for crosstalk tests.

   (c) C not required for any other tests.

FIGURE 7b. Logic load circuit.

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* TABLE XI. Output voltage, low level, test.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
K1	DATA BUS 0	98	12.0	0.0	0.4	V	1-5
K2	DATA BUS 1	49					
K3	DATA BUS 2	96					
K4	DATA BUS 3	100					
K5	DATA BUS 4	99					
K6	DATA BUS 5	50					
K7	DATA BUS 6	48					
K8	DATA BUS 7	46					
K9	DATA BUS 8	94					
K10	DATA BUS 9	45					
K11	DATA BUS 10	93					
K12	DATA BUS 11	97					
K13	DATA BUS 12	95					
K14	DATA BUS 13	47					
K15	DATA BUS 14	43					
K16	DATA BUS 15	42					
K17	DISCRETE 0	35					
K18	DISCRETE 1	81					
K19	DISCRETE 2	84					
K20	DISCRETE 3	32	12.0	0.0	0.4	V	1-5

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* TABLE XI. Output voltage, low level, test - Continued.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
K21	DISCRETE 4	78	12.0	0.0	0.4	V	1-5
K22	DISCRETE 5	28					
K23	DISCRETE 6	75					
K24	DISCRETE 7	24					
K25	-DRVR ENBL 0	82					
K26	-DRVR ENBL 1	86					
K27	-DRVR ENBL 2	37					
K28	-DRVR ENBL 3	36					
K29	-DRVR ENBL 4	73					
K30	-DRVR ENBL 5	79					
K31	-DRVR ENBL 6	77					
K32	-DRVR ENBL 7	29	12.0				
K33	-IE CMD REQ	70	4.0				
K34	-IE DATA REQ	69					
K35	-OE REQ	19					
K36	-ERROR REQ	20	4.0				
K37	DIFF A0 OUT	16	40.0				
K38	DIFF B0 OUT	64					
K39	DIFF A1 OUT	14					
K40	DIFF B1 OUT	63	40.0	0.0	0.4	V	1-5

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* TABLE XI. Output voltage, low level, test - Continued.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
K41	-RCVR 0 ENBL	59	12.0	0.0	0.4	V	1-5
K42	-RCVR 1 ENBL	62					
K43	-RCVR ENBL	8					
K44	-LINK NOT BSY	60					
K45	-DREQ	9					
K46	-DATA XFR STB	12					
K47	SEND CLOCK	54					
K48	-DRVR 1 DATA OUT	52					
K49	-BIPLR 1 OUT	11	12.0				
K50	-DRVR DATA OUT	55	10.4				
K51	-RCVR 1 DATA	65	16.0				
K52	-RCVR DATA	66	11.7	0.0			
K53	-BIAS	76	0.1	0.26	0.4	V	1-5

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NOTES:

1. Output voltage, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V	± 45 mV	K1-K53
Power supply current	1.058 A	NA	K1-K53
Forcing current, (Iol)	12.0 µA 4.0 µA 40.0 µA 12.0 µA 10.4 µA 16.0 µA 11.7 µA 0.1 µA	± 120 µA ± 40 µA ± 400 µA ± 120 µA ± 104 µA ± 160 µA ± 112 µA ± 1 µA	K1-K32 K33-K36 K37-K40 K41-K49 K50 K51 K52 K53
* Pattern voltage high, (VH) low, (VL)	 2.0V, 2.2V 0.8V, 0.7V	 ± 20 mV, ± 22 mV ± 8 mV, ± 7 mV	 K1-K53 K1-K53
Output voltage measurement	NA	± 20 mV	K1-K53

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing current shall be applied to the pin specified in the appropriate test conditions in table XI and the voltage then measured.

3. The load shall be as indicated on figure 7.

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NOTES: - Continued.

4. Abbreviations used are defined in 6.1.

*5. For pattern voltage, the high and low level input threshold voltage for each pin is as specified in 6.3.1.

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TABLE XII. Output leakage current, high level, test.

Test cond	Signal name	Pin no.	Forcing voltage (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
L1	DATA BUS 0	98	2.7	-20	20	µA	1-3
L2	DATA BUS 1	49					
L3	DATA BUS 2	96					
L4	DATA BUS 3	100					
L5	DATA BUS 4	99					
L6	DATA BUS 5	50					
L7	DATA BUS 6	48					
L8	DATA BUS 7	46					
L9	DATA BUS 8	94					
L10	DATA BUS 9	45					
L11	DATA BUS 10	93					
L12	DATA BUS 11	97					
L13	DATA BUS 12	95					
L14	DATA BUS 13	47					
L15	DATA BUS 14	43					
L16	DATA BUS 15	42	2.7	-20	20		
L17	-IE CMD REQ	70	5.5	-100	100		
L18	-IE DATA REQ	69					
L19	-OE REQ	19					
L20	-ERROR REQ	20		-100	100		
L21	-RCVR 1 DATA	65	5.5	-200	200	µA	1-3

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NOTES:

1. Output leakage current, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	L1-L21
Power supply current limit	1.058 A	NA	L1-L21
Forcing voltage, (Voh)	2.7 V 5.5 V	± 135 mV ± 55 mV	L1-L16 L17-L21
Pattern voltage high, (VH) low, (VL)	3.0 V 0.0 V	± 100 mV ± 100 mV	L1-L21 L1-L21
Output current measurement	NA | NA	± 2 µA ± 10 µA ± 20 µA	L1-L16 L17-L20 L21

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

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NOTES: - Continued.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table XII and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE XIII. Output leakage current, low level, test.

Test cond	Signal name	Pin no.	Vol (V)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
M1	DATA BUS 0	98	0.4	-200	20	µA	1-3
M2	DATA BUS 1	49					
M3	DATA BUS 2	96					
M4	DATA BUS 3	100					
M5	DATA BUS 4	99					
M6	DATA BUS 5	50					
M7	DATA BUS 6	48					
M8	DATA BUS 7	46					
M9	DATA BUS 8	94					
M10	DATA BUS 9	45					
M11	DATA BUS 10	93					
M12	DATA BUS 11	97					
M13	DATA BUS 12	95					
M14	DATA BUS 13	47					
M15	DATA BUS 14	43					
M16	DATA BUS 15	42	0.4	-200	20	µA	1-3

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NOTES:

1. Output leakage current, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	M1-M16
Power supply current limit	1.058 A	NA	M1-M16
Forcing voltage, (Vol)	0.4 V	± 20 mV	M1-M16
Pattern voltage high, (VH)	3.0 V	± 100 mV	M1-M16
low, (VL)	0.0 V	± 100 mV	M1-M16
Output current measurement	NA	± 11 μA	M1-M16

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table XIII and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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* TABLE XIV. Power supply current test.

Test cond	Parameter	Pin no.	25 ± 5°C Initial limits		Units	Notes
			Min	Max		
N	Icc	87	35	705	mA	1-4

NOTES:

1. Power supply current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	N
Power supply current limit	1.058 A	NA	N
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 N N
Power supply current measurement	NA	± 33.5 mA	N

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test and take a measurement at the pattern line with the maximum power supply current as indicated in table XIV.

3. Abbreviations used are defined in 6.1.

*4. Pins 30 (MASTER/-SLAVE) and 74 (-DISABLE PORT CMDS) shall be tied to 0.0V during this test.

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TABLE XV. Functional test.

Test cond	Test description	Notes
P1	Functional test minimum power supply voltage maximum pattern rate	1-3
P2	Functional test maximum power supply voltage maximum pattern rate	1-3

NOTES:

1. Functional test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V 5.5 V	± 45 mV ± 55 mV	P1 P2
Power supply current limit	1.058 A	NA	P1,P2
* Pattern voltage high, (VH) low, (VL)	 2.4 V 0.4 V	 ± 120 mV ± 20 mV	 P1,P2 P1,P2
Output voltage verification high, (Voh) high, (Voh) low, (Vol)	 NA | NA	 ± 105 mV ± 155 mV ± 25 mV	 P1 P2 P1,P2

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Use the appropriate pattern to verify the functional performance of the module as indicated in table XV using only a capacitive load of 100 pF (see figure 7). The output compare levels shall be as follows:

High level: $2.4\ V \leq Voh \leq Vcc$
Low level: $0.0\ V \leq Vol \leq 0.4\ V$

(c) The module inputs shall not violate the minimum input timing parameters referenced on figure 8.

3. Abbreviations used are defined in 6.1.

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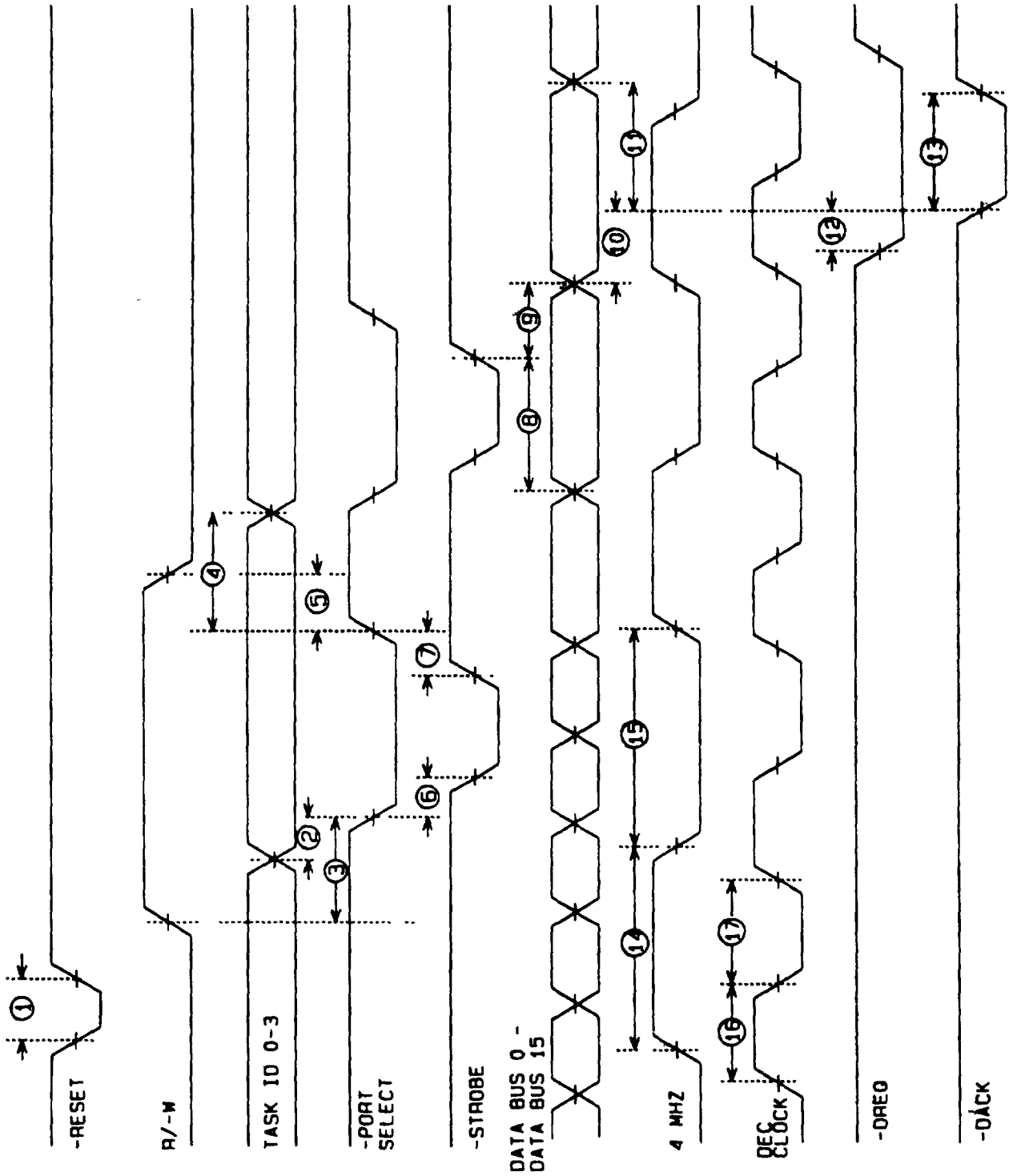


FIGURE 8. Input timing parameters for table XV.

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NOTES:

1. Input parameters:

1	= 90 nS min.
2	= 0 nS min.
3	= 0 nS min.
4	= 0 nS min.
5	= 0 nS min.
6	= 0 nS min.
7	= 20 nS min.
8	= 30 nS min.
9	= 80 nS min.
10	= 30 nS min.
11	= 80 nS min.
12	= 0 nS min.
13	= 175 nS min.
14	= 120 nS min.
15	= 120 nS min.
16	= 40 nS min.
17	= 40 nS min.

2. Input parameters are referenced at the 1.5 V level.

3. Pattern voltages shall be specified in the test tolerance table.

4. Transition times (as measured between the 0.7 V and 2.0 V points) shall not exceed 20 nS.

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TABLE XVI. Dynamic test.

Test cond	Test description	Notes
Q1	Dynamic test, sample maximum pattern rate minimum power supply	1-7
Q2	Dynamic test, sample maximum pattern rate maximum power supply	1-7
Q3	Dynamic test, qualification maximum pattern rate minimum power supply	1-7
Q4	Dynamic test, qualification maximum pattern rate maximum power supply	1-7

NOTES:

1. Dynamic test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V 5.5 V	± 45 mV ± 55 mV	Q1,Q3 Q2,Q4
Power supply current limit	1.058 A	NA	Q1-Q4
Input voltage high, (VH) low, (VL)	 2.4 V 0.4 V	 ± 120 mV ± 20 mV	 Q1-Q4 Q1-Q4

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Output voltage verification			
high, (Voh)	NA	± 105 mV	Q1,Q3
high, (Voh)	|	± 155 mV	Q2,Q4
low, (Vol)	NA	± 25 mV	Q1-Q4

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) The output compare levels shall be as follows:

High level: $Vref < Voh \leq Vcc$
Low level: $0.0\ V \leq Vol \leq Vref$

3. The load circuit for this test is specified on figure 7.

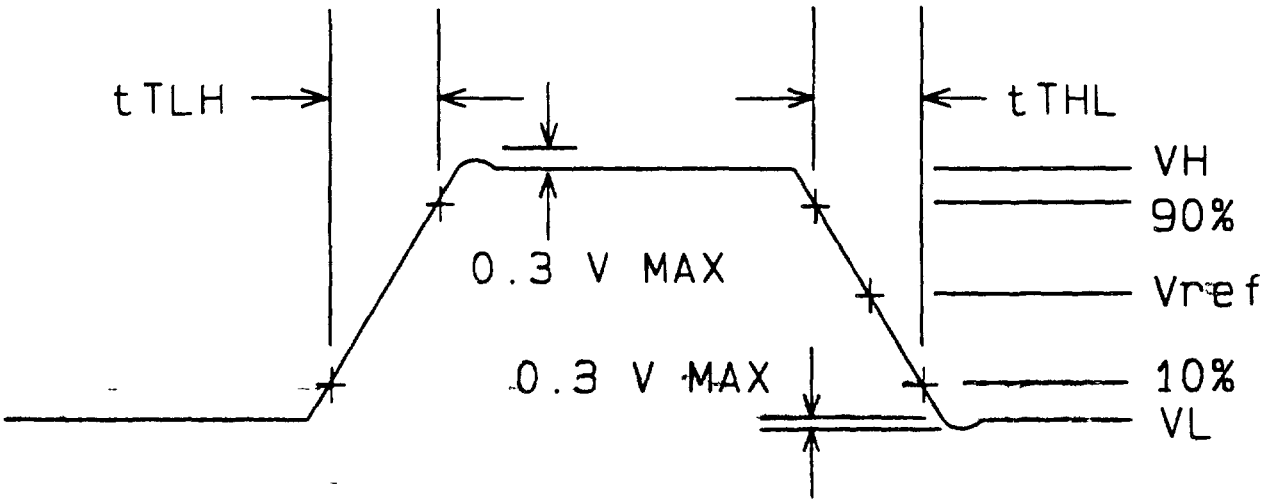
4. All of the input waveforms shall meet the requirements shown on figure 9.

5. This test shall be performed as indicated in table XVI. The dynamic parameters are found in table XVII.

6. I/O waveforms are as referenced on figure 10.

7. Abbreviations used are defined in 6.1.

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NOTES:

1. VH and VL are as specified in the test tolerance table.
2. Vref = 1.5 V.
3. Rise and fall times for the above waveform are as follows:

Input pin	tTLH	tTHL
98,49,96,100,99,50,48,46,94, 45,93,97,95,47,43,42,87,33, 41,38,91,92,44,90,83,26,34, 80,85,39,89,88,5,53,51,4,6,3, 2,56,7,61,58,17,71,57	15 ± 1.5 nS	15 ± 1.5 nS
21,18,23,67,13	5 ± 1.5 nS	5 ± 1.5 nS
30,74	20 ± 1.5 nS	20 ± 1.5 nS

FIGURE 9. Sample input waveform for table XVI.

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TABLE XVII Dynamic parameters

Ref num	Parameter number	Parameter	Minimum limit			Maximum limit			Units	Fig
			0°C	25°C	60°C	0°C	25°C	60°C		
2	1	-STROBE to -OE REQ, tPLH	-	-	-	229	204	229	nS	10a
5	2	-STROBE to -OE REQ, tPHL				224	200	224		
12	3	4MHZ to -OE REQ, tPLH				239	214	239		
10	4	4MHZ to -OE REQ, tPHL				234	210	234		
8	5	-RESET to -OE REQ, tPLH				229	204	229		
1	6	-STROBE to -IE CMD REQ, tPLH				229	204	229		
4	7	-STROBE to -IE CMD REQ, tPHL				224	200	224		
14	8	4MHZ to -IE CMD REQ, tPLH				239	214	239		
11	9	4MHZ to -IE CMD REQ, tPHL				234	210	234		
7	10	-RESET to -IE CMD REQ, tPLH				179	154	179		
1	11	-STROBE to -IE DATA REQ, tPLH				229	204	229		
4	12	-STROBE to -IE DATA REQ, tPHL				224	200	224		
14	13	4MHZ to -IE DATA REQ, tPLH				239	214	239		
11	14	4MHZ to -IE DATA REQ, tPHL				234	210	234		
7	15	-RESET to -IE DATA REQ, tPLH				179	154	179		
3	16	-STROBE to -ERROR REQ, tPLH	-	-	-	229	204	229	nS	10a

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TABLE XVII Dynamic parameters - Continued.

Ref num	Parameter number	Parameter	Minimum limit			Maximum limit			Units	Fig
			0°C	25°C	60°C	0°C	25°C	60°C		
6	17	-STROBE to -ERROR REQ, tPHL	-	-	-	224	200	224	nS	10a
15	18	4MHZ to -ERROR REQ, tPLH				239	214	239		
9	20	-RESET to -ERROR REQ, tPLH				179	154	179		10a
18	21	DEC CLOCK to -LINK NOT BSY, tPLH				163	154	163		10b
16	22	DEC CLOCK to -LINK NOT BSY, tPHL				168	158	168		
27	23	4MHZ to -LINK NOT BSY, tPHL				118	108	118		
22	24	-RESET to -LINK NOT BSY, tPLH				158	148	158		
23	25	DEC CLOCK to SEND CLOCK, tPLH				103	93	103		
19	26	DEC CLOCK to SEND CLOCK, tPHL				103	93	103		
21	27	4MHZ to -DATA XFR STB, tPLH				196	186	196		
20	28	4MHZ to -DATA XFR STB, tPHL				186	178	186		
26	29	-RESET to -DATA XFR STB, tPLH				186	178	186		10b
29	30	-DACK to -DREQ, tPLH				103	93	103		10c
28	31	4MHZ to -DREQ, tPHL				113	103	113		
33	32	-RESET to -DREQ, tPLH				128	118	128		10c
24	33	DEC CLOCK to -DRVR DATA OUT, tPLH	-	-	-	153	143	153	nS	10b

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TABLE XVII Dynamic parameters - Continued.

Ref num -	Parameter number	Parameter	Minimum limit			Maximum limit			Units	Fig
			0°C	25°C	60°C	0°C	25°C	60°C		
17	34	DEC CLOCK to -DRVR DATA OUT, tPHL	-	-	-	153	143	153	nS	10b
36	35	DEC CLOCK to DIFF A0 OUT, tPLH				187	177	187		10d
43	36	DEC CLOCK to DIFF A0 OUT, tPHL				182	172	182		
44	37	DEC CLOCK to DIFF B0 OUT, tPLH				187	177	187		
37	38	DEC CLOCK to DIFF B0 OUT, tPHL				182	172	182		
45	39	-ENABLE 0 IN to DIFF A0 OUT, tPLH				69	59	69		
47	40	-ENABLE 0 IN to DIFF A0 OUT, tPHL				59	50	59		
48	41	-ENABLE 0 IN to DIFF B0 OUT, tPLH				69	59	69		
46	42	-ENABLE 0 IN to DIFF B0 OUT, tPHL				59	50	59		
49	43	-DRVR 1 DATA IN to DIFF A1 OUT, tPLH				41	36	41		
41	44	-DRVR 1 DATA IN to DIFF A1 OUT, tPHL								
42	45	-DRVR 1 DATA IN to DIFF B1 OUT, tPLH	-	-	-	41	36	41	nS	10d

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TABLE XVII Dynamic parameters - Continued.

Ref num	Parameter number	Parameter	Minimum limit			Maximum limit			Units	Fig
			0°C	25°C	60°C	0°C	25°C	60°C		
50	46	-DRVR 1 DATA IN to DIFF B1 OUT, tPHL	-	-	-	41	36	41	nS	10d
51	47	DIFF A0 IN to -RCVR DATA, tPLH				91	66	91		10e
52	48	DIFF A0 IN to -RCVR DATA, tPHL								
51	49	DIFF B0 IN to -RCVR DATA, tPLH								
52	50	DIFF B0 IN to -RCVR DATA, tPHL				91	66	91		
53	51	-RCVR ENBL 0 to -RCVR DATA, tPLH				98	71	98		
54	52	-RCVR ENBL 0 to -RCVR DATA, tPHL				96	67	96		
51	53	DIFF A1 IN to -RCVR 1 DATA, tPLH				91	66	91		
52	54	DIFF A1 IN to -RCVR 1 DATA, tPHL								
51	55	DIFF B1 IN to -RCVR 1 DATA, tPLH								
52	56	DIFF B1 IN to -RCVR 1 DATA, tPHL				91	66	91		
53	57	-RCVR ENBL 1 to -RCVR 1 DATA, tPLH				98	71	98		
54	58	-RCVR ENBL 1 to -RCVR 1 DATA, tPHL				96	67	96		10e
59	59-66	4MHZ to DISCRETE 0-7, tPLH				138	129	138		10f
56	67-74	4MHZ to DISCRETE 0-7, tPHL	-	-	-	143	133	143	nS	10f

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TABLE XVII Dynamic parameters - Continued

Ref num	Parameter number	Parameter	Minimum limit			Maximum limit			Units	Fig
			0°C	25°C	60°C	0°C	25°C	60°C		
62	75-82	-RESET to DISCRETE 0-7, tPHL	-	-	-	128	118	128	nS	10f
24	83	DEC CLOCK to -BIPLR 1 OUT, tPLH	|	|	|	216	206	216	|	10b
17	84	DEC CLOCK to -BIPLR 1 OUT, tPHL	-	-	-	206	197	206	|	10b
57	85-100	DATA BUS0 - DATA BUS15 4MHZ, setup time, tS	29	29	29	-	-	-	|	10f
58	101-116	DATA BUS0 - DATA BUS15 4MHZ, hold time, tH	79	79	79	|	|	|	|	|
60	117-132	STATUS 0 - STATUS 15 to 4MHZ, setup time, tS	29	29	29	|	|	|	|	|
61	133-148	STATUS 0- STATUS 15 to 4MHZ, hold time, tH	79	79	79	-	-	-	|	10f
67	149	-STROBE to -DRVR 1 DATA OUT, tPLH	-	-	-	158	148	158	|	10g
71	150	-STROBE to -DRVR 1 DATA OUT, tPHL	|	|	|	158	148	158	|	10g
24	151	DEC CLOCK to -DRVR 1 DATA OUT, tPLH	|	|	|	193	183	193	|	10b
17	152	DEC CLOCK to -DRVR 1 DATA OUT, tPHL	|	|	|	193	183	193	|	10b
72	153	-STROBE to -RCVR 0 ENBL, tPLH	-	-	-	158	144	158	nS	10g

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TABLE XVII Dynamic parameters - Continued.

Ref num	Parameter number	Parameter	Minimum limit			Maximum limit			Units	Fig
			0°C	25°C	60°C	0°C	25°C	60°C		
68	154	-STROBE to -RCVR 0 ENBL, tPHL	-	-	-	158	148	158	nS	10g
67	155	-STROBE to -RCVR 1 ENBL, tPLH					144			
71	156	-STROBE to -RCVR 1 ENBL, tPHL				158	148	158		
70	157	-RESET to -RCVR 0 ENBL, tPHL				128	118	128		
69	158	-RESET to -RCVR 1 ENBL, tPLH				128	118	128		
72	159	-STROBE to -RCVR ENBL, tPLH				158	144	158		
68	160	-STROBE to -RCVR ENBL, tPHL				158	148	158		
70	161	-RESET to -RCVR ENBL, tPHL				128	118	128		
67, 72	162-169	-STROBE to -DRVR ENBL 0-7, tPLH				188	174	188		
68, 71	170-177	-STROBE to -DRVR ENBL 0-7, tPHL				188	178	188		
70, 69	178	-RESET to -DRVR ENBL 0, tPHL				168	158	168		
69, 70	179-185	-RESET to -DRVR ENBL 1-7, tPLH				168	154	168		10g
63	186-201	STATUS 0-15 to DATA BUS 0-15, tPLH	-	-	-	57	47	57	nS	10i

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Parameter number	Parameter	Minimum limit			Maximum limit			Units	Fig
			0°C	25°C	60°C	0°C	25°C	60°C		
63	202-217	STATUS 0-15 to DATA BUS 0-15, tPHL	-	-	-	57	47	57	nS	10f
79	218-233	-STROBE to DATA BUS 0-15, tPZH				163	145	163		10h
79	234-249	-STROBE to DATA BUS 0-15, tPZL				163	145	163		10h
35	250-265	-DACK to DATA BUS 0-15, tPZH				251	246	251		10c
35	266-281	-DACK to DATA BUS 0-15, tPZL	-	-	-	251	246	251		
31	282-297	DATA BUS 0-15 to -DACK, setup time, tS	29	29	29	-	-	-		
32	298-313	DATA BUS 0-15 to -DACK, hold time, tH	89	89	89					10c
83	314-329	DATA BUS 0-15 to -STROBE, setup time, tS	29	29	29					10h
84	330-345	DATA BUS 0-15 to -STROBE, hold time, tH	79	79	79					
77	346	-PORT SELECT pulse width low, PWL	110	110	110					
78	347	-STROBE pulse width low, PWL	90	90	90					
75	348	R/-W to -PORT SELECT, setup time, tS	0	0	0	-	-	-	nS	10h

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TABLE XVII Dynamic parameters - Continued.

Ref num	Parameter number	Parameter	Minimum limit			Maximum limit			Units	Fig
			0°C	25°C	60°C	0°C	25°C	60°C		
81	349	R/-W to -PORT SELECT, hold time, tH	0	0	0	-	-	-	nS	10h
74	350-353	TASK ID 0-3 to -PORT SELECT, setup time, tS								
82	354-357	TASK ID 0-3 to -PORT SELECT, hold time, tH								
76	358	-PORT SELECT to -STROBE, setup time, tS	0	0	0					
80	359	-PORT SELECT to -STROBE, hold time, tH	20	20	20	-	-	-		10h
65	360	4MHZ PWL	120	120	120	130	130	130		10f
64	361	4MHZ PWH	120	120	120	130	130	130	nS	
66	362	4MHZ Fmax	-	-	-	4	4	4	MHz	10f
40	363	DEC CLOCK PWL	40	40	40	45	45	45	nS	10d
38	364	DEC CLOCK PWH	40	40	40	45	45	45	nS	
39	365	DEC CLOCK Fmax	-	-	-	12	12	12	MHz	10d
34	366	-RESET PWL	90	90	90	-	-	-	nS	10c
30	367	-DACK PWL	175	175	175					10c
55	368-383	4MHZ to DATA BUS 0-15, tPZH	238	220	238	-	-	-	nS	10f

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TABLE XVII. Dynamic parameters - Continued

Ref num	Parameter number	Parameter	Minimum limit			Maximum limit			Units	Fig
			0°C	25°C	60°C	0°C	25°C	60°C		
55	384-399	4MHZ to DATA BUS 0-15, tPZL	238	220	238	-	-	-	nS	10f
79	400-415	-STROBE to DATA BUS 0-15, tPZH	-	-	-	180	162	180		10h
79	416-431	-STROBE to DATA BUS 0-15, tPZL				180	162	180		10h
35	432-447	-DACK to DATA BUS 0-15, tPZH				268	263	268		10c
35	448-463	-DACK to DATA BUS 0-15, tPZL				268	263	268		10c
55	464-479	4MHZ to DATA BUS 0-15, tPZH				255	237	255		10f
55	480-495	4MHZ to DATA BUS 0-15, tPZL	-	-	-	255	237	255	nS	10f

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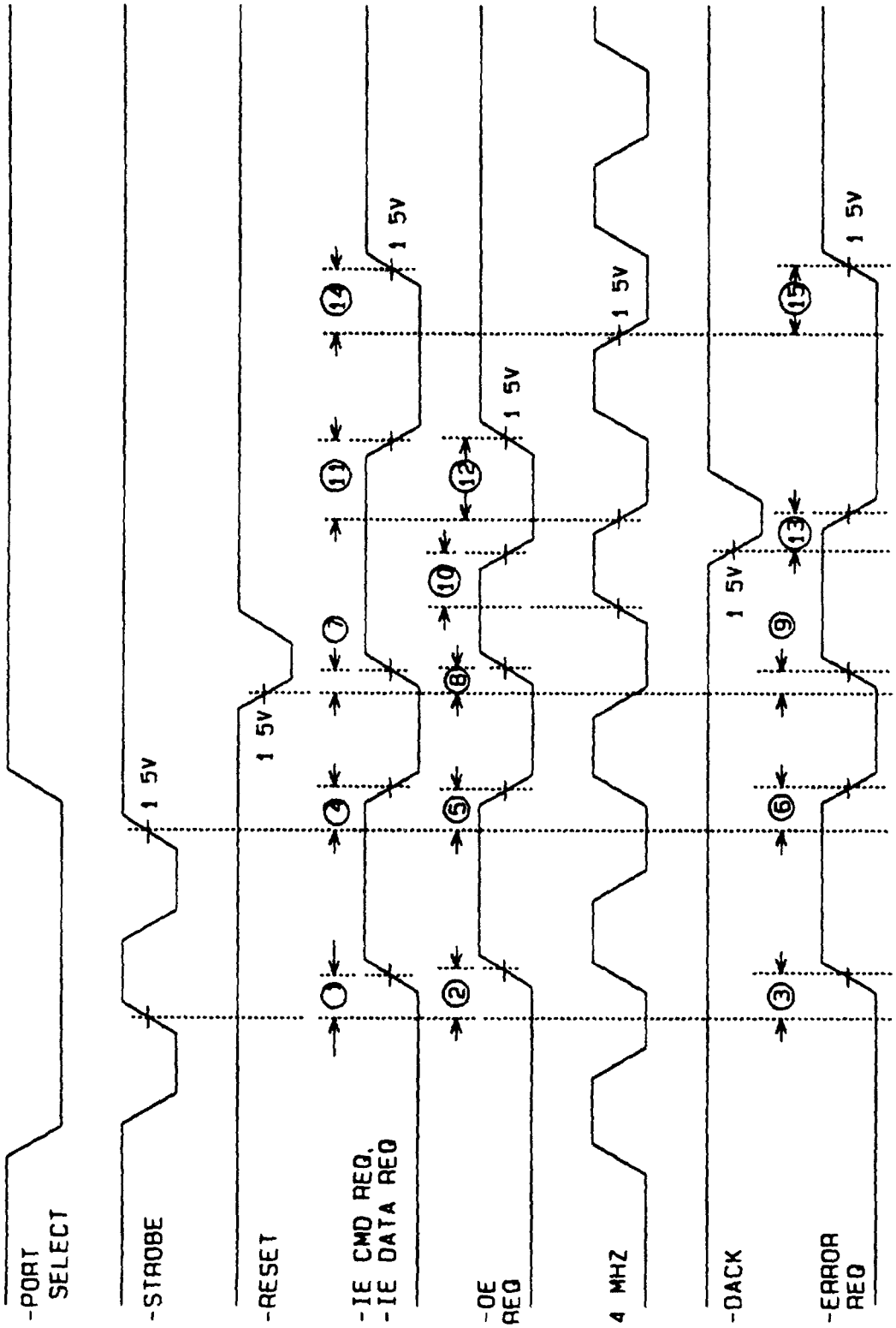


FIGURE 10a. Dynamic waveforms for table XVII.

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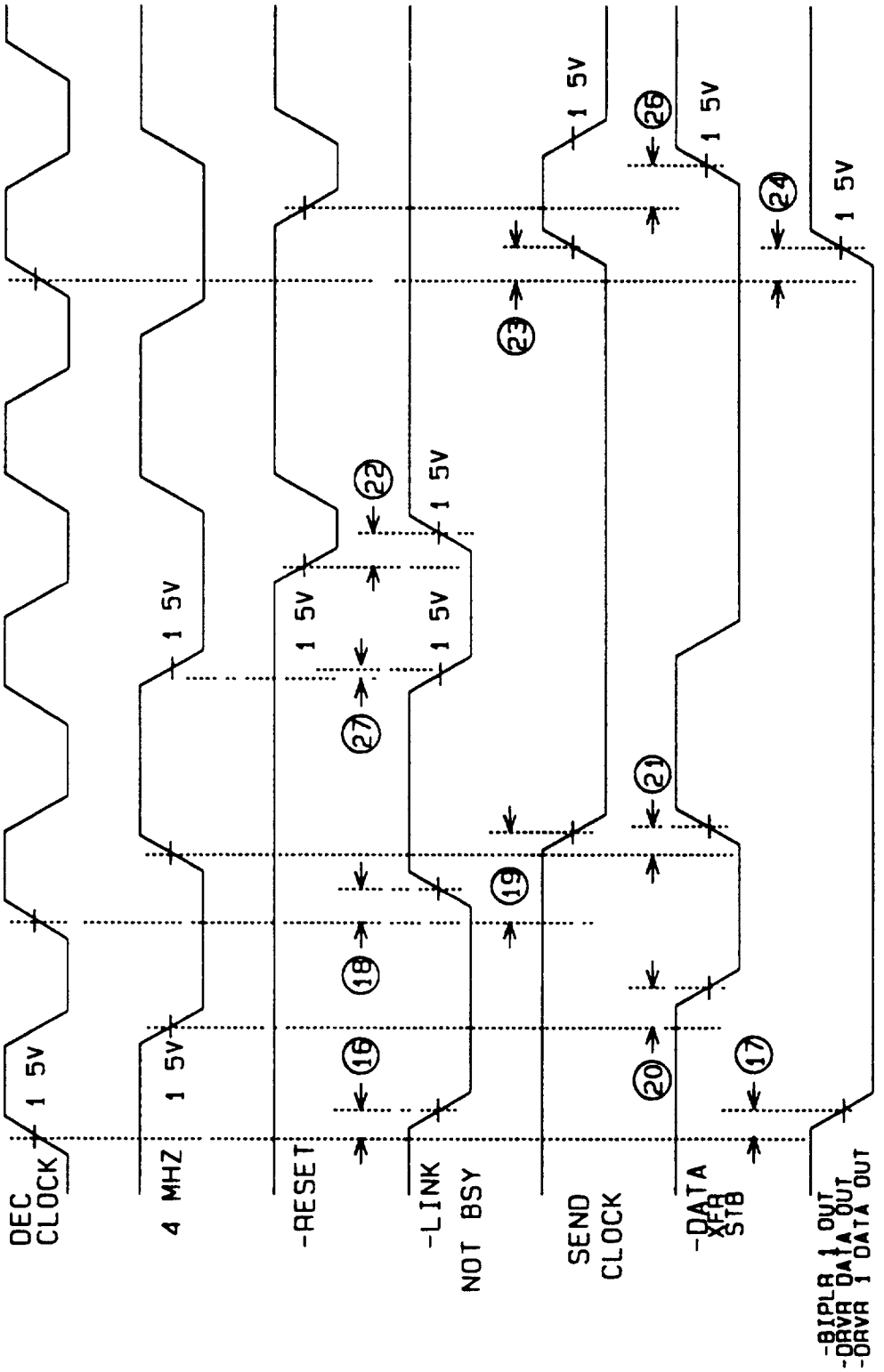


FIGURE 10b. Dynamic waveforms for table XVII.

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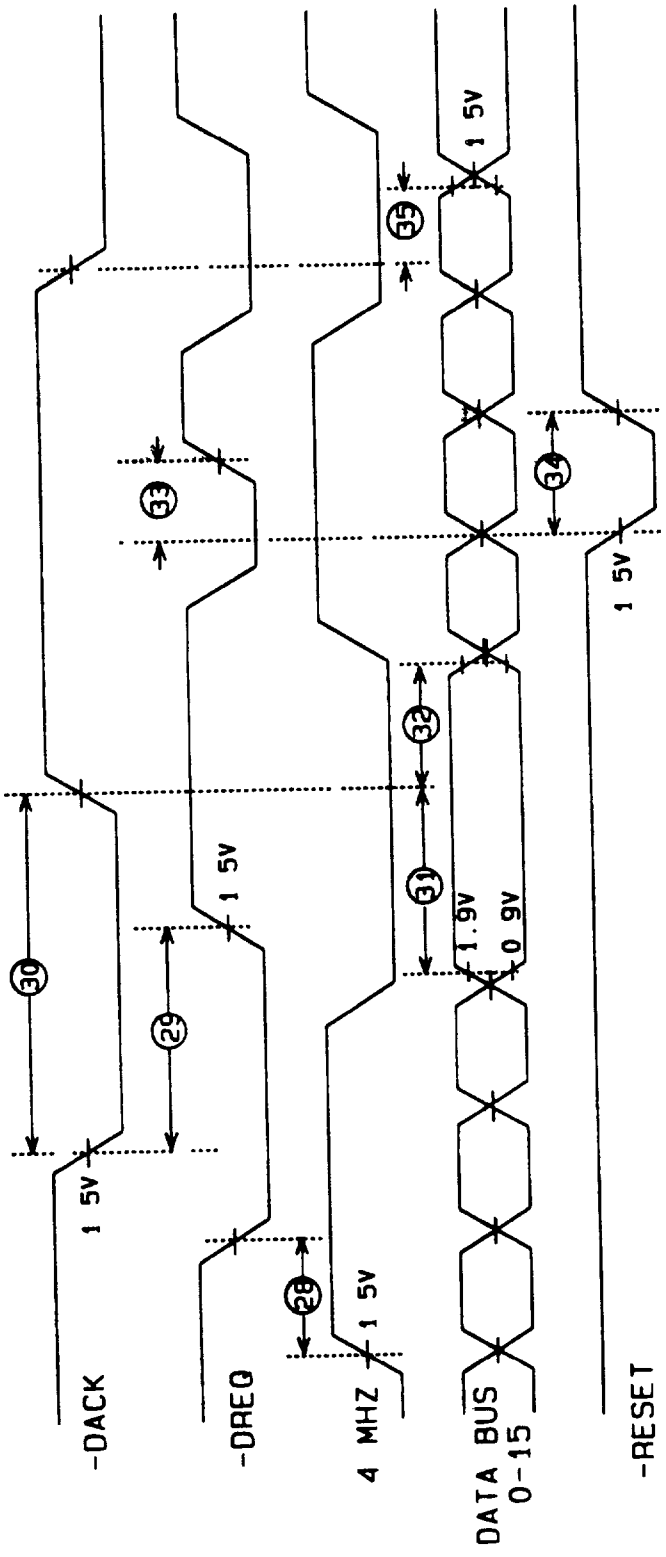


FIGURE 10c. Dynamic waveforms for table XVII.

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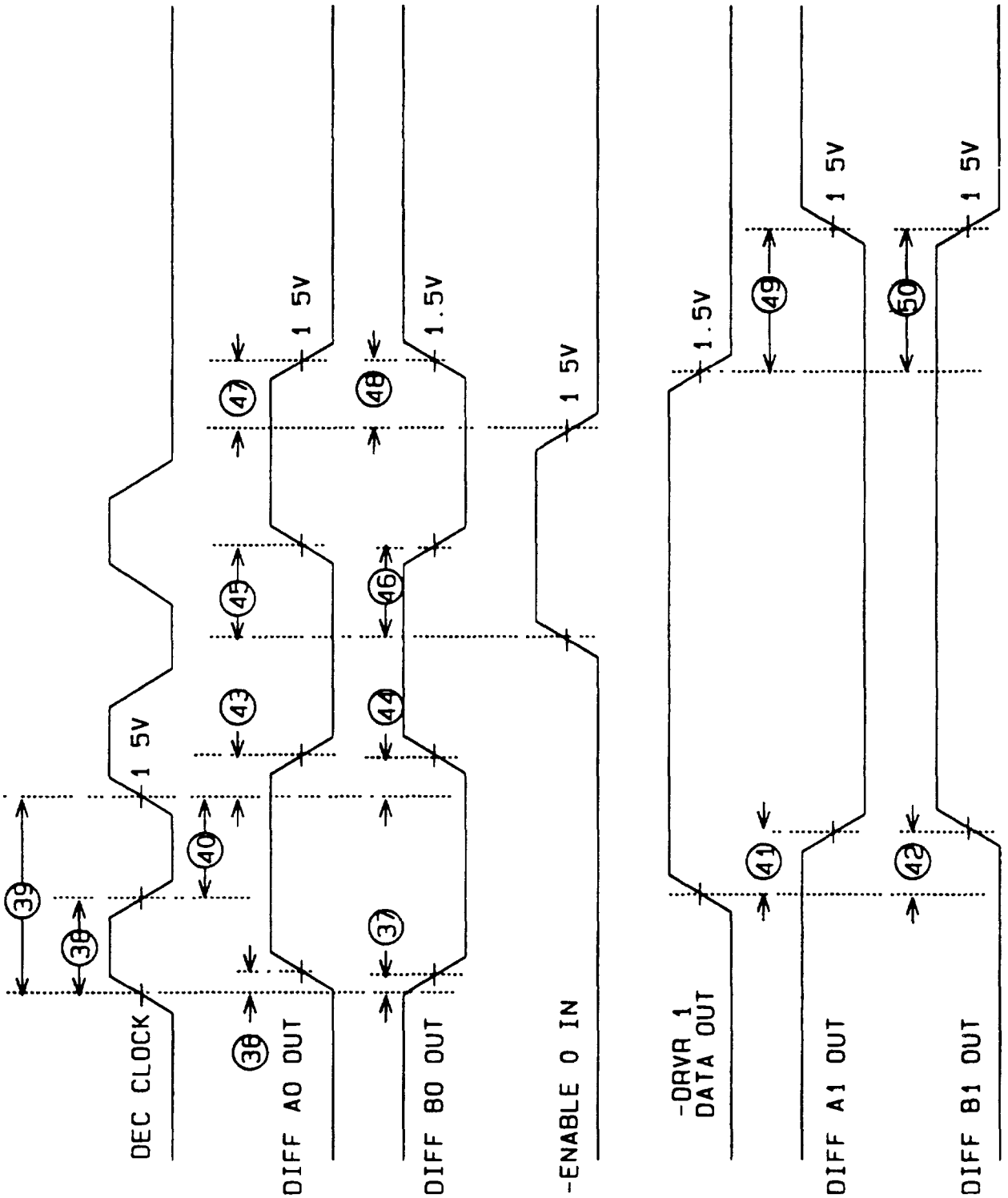


FIGURE 10d. Dynamic waveforms for table XVII.

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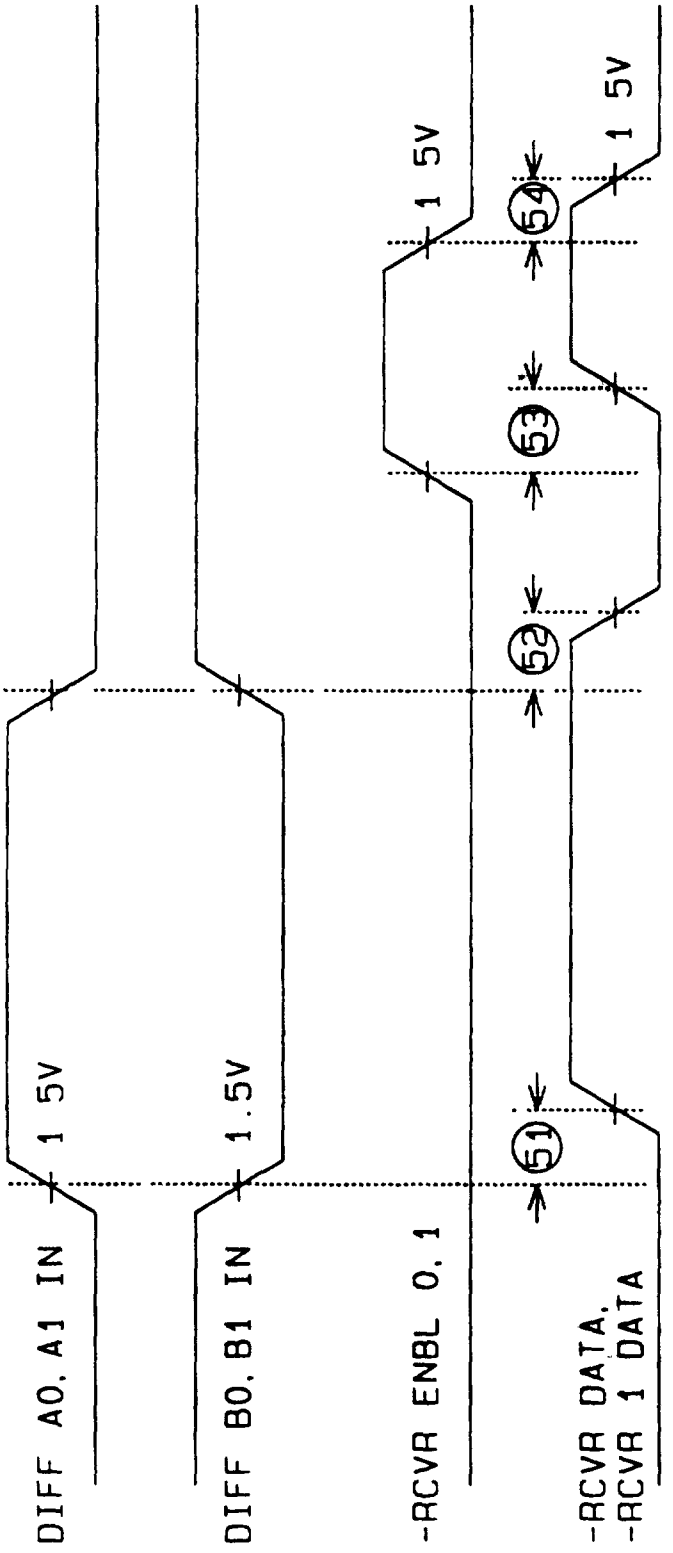


FIGURE 10e. Dynamic waveforms for table XVII.

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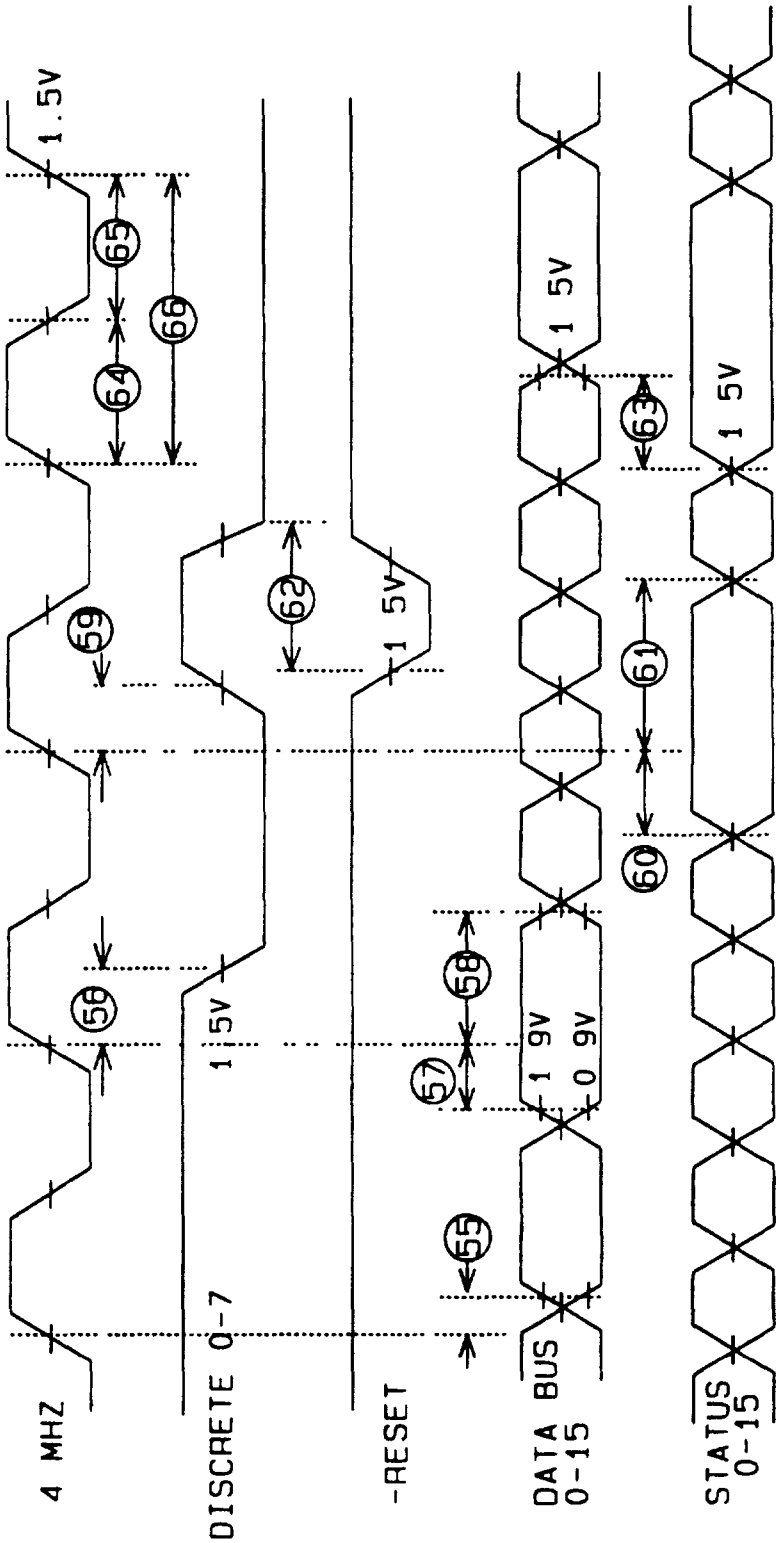


FIGURE 10f. Dynamic waveforms for table XVII.

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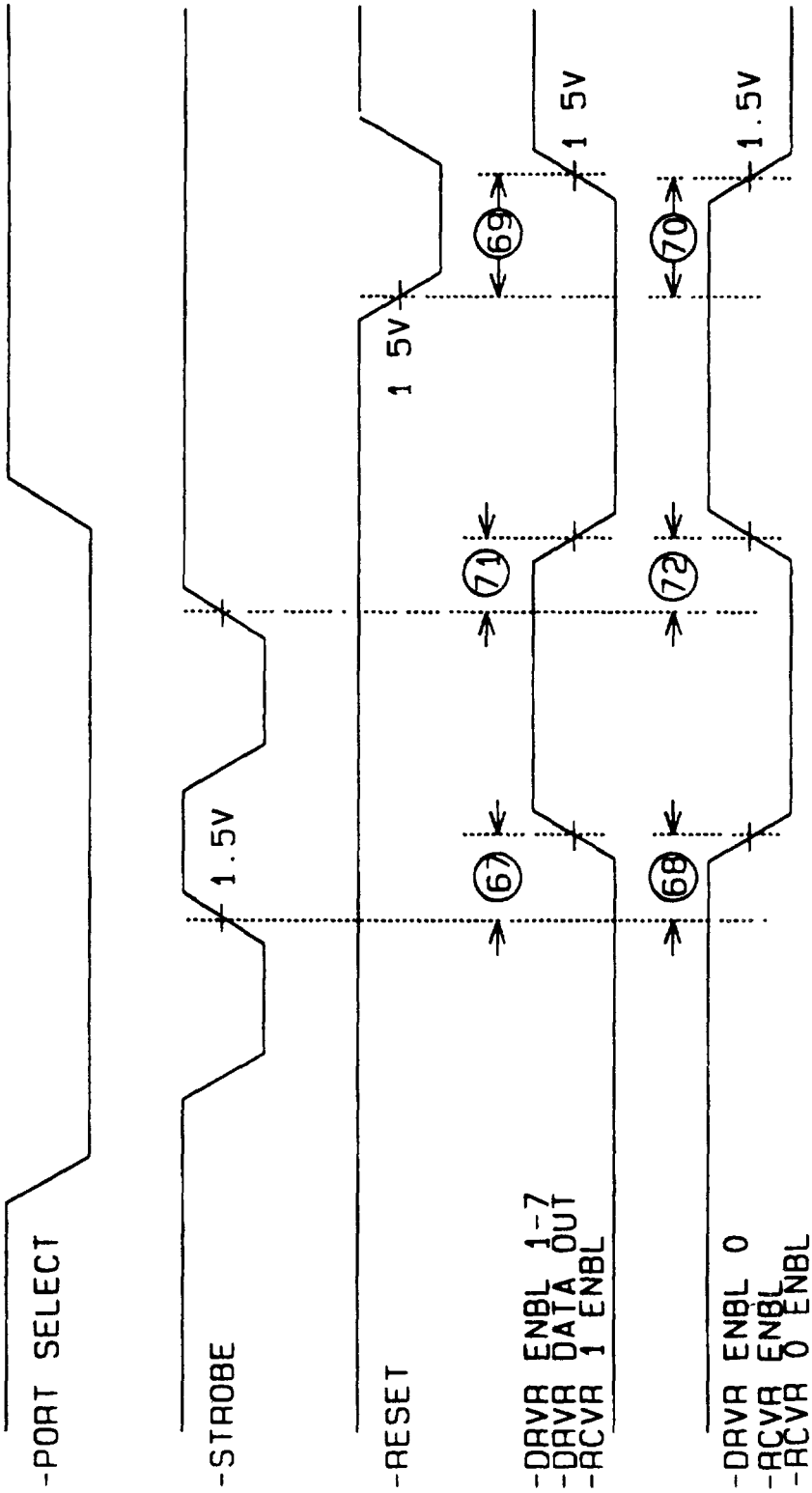


FIGURE 10g. Dynamic waveforms for table XVII.

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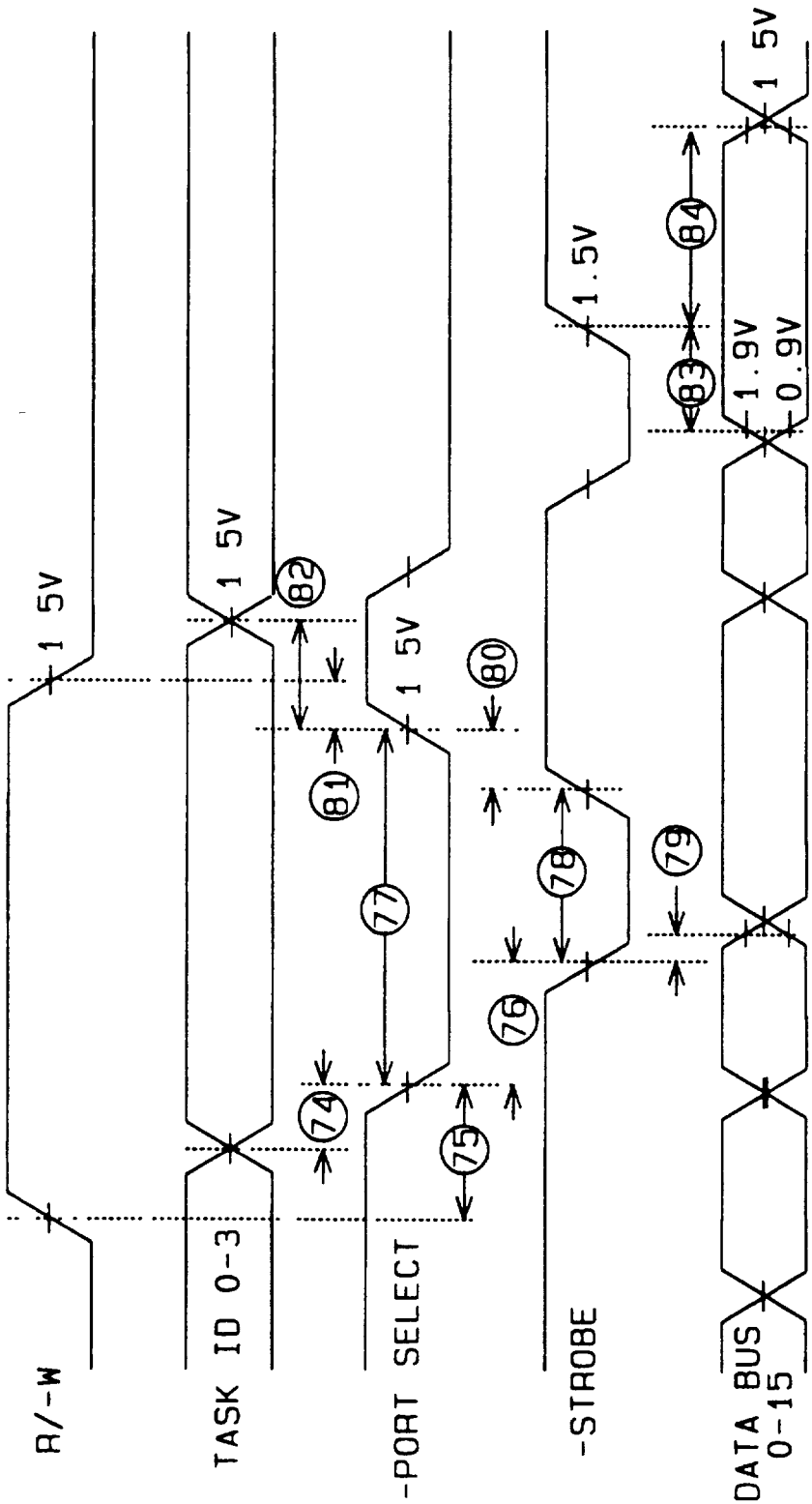


FIGURE 10h. Dynamic waveforms for table XVII.

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TABLE XVIII. Power supply filter capacitance test.

Test cond	Parameter	Pin numbers	25 ± 5°C Initial limits		Units	Notes
			Min	Max		
R	PSFC	1,31	8.1	12.3	μF	1-3

NOTES:

1. Power supply filter capacitance test tolerance table.

Condition	Value	Accuracy	Test condition
Capacitance bridge frequency	1 KHz	± 100 Hz	R
Capacitance bridge voltage	0.3 Vp	± 30 mVp	R
Capacitance measurement	NA	± 210 nF	R

2. The capacitance bridge used in this test shall be item C01. The negative terminal shall be connected to the module ground and the positive terminal to the pin indicated in table XVIII.

3. Abbreviations used are defined in 6.1.

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TABLE XIX. Power supply transient amplitude test.

Test cond	Parameter	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
S	PSTA	1,31	0	400	mV	1-4

NOTES:

1. Power supply transient amplitude test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.0 V	± 250 mV	S
Power supply current limit	1.058 A	NA	S
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 S S
Power supply transient amplitude measurement	NA	± 20 mV	S

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table to an external source. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test and make the indicated measurement for the test condition shown in table XIX.

(c) All Vcc pins shall be shorted together and the scope point, probe 'A', shall be connected to the pin indicated in table XIX.

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NOTES: - Continued.

(d) Using the oscilloscope, D06, inspect for the most negative spike as shown on figure 11 at the pattern line with the maximum power supply current.

(e) The maximum probe capacitance is 15 pF. There shall be no other capacitance present on Vcc to ground.

3. The module inputs shall not violate the minimum input timing requirements shown on figure 8.

4. Abbreviations used are defined in 6.1.

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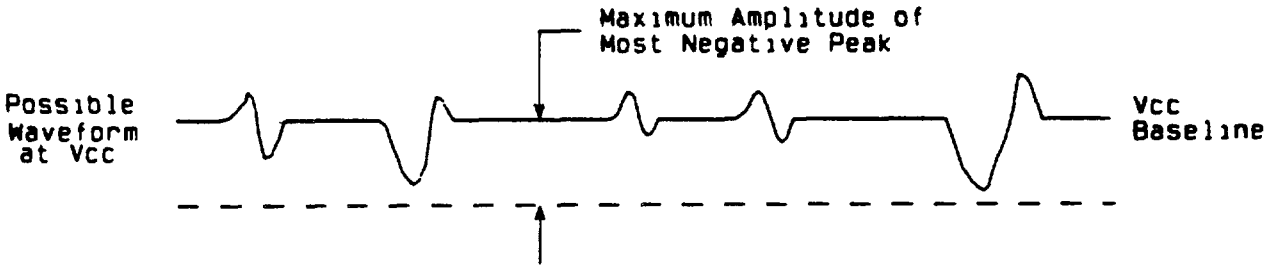


FIGURE 11. Power supply transient amplitude waveform for table XIX.

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* TABLE XX. Output crosstalk, high level, test.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes	Figs
T1	DATA BUS 0	98	0	1100	mV	1-4	12,13
T2	DATA BUS 1	49					
T3	DATA BUS 2	96					
T4	DATA BUS 3	100					
T5	DATA BUS 4	99					
T6	DATA BUS 5	50					
T7	DATA BUS 6	48					
T8	DATA BUS 7	46					
T9	DATA BUS 8	94					
T10	DATA BUS 9	45					
T11	DATA BUS 10	93					
T12	DATA BUS 11	97					
T13	DATA BUS 12	95					
T14	DATA BUS 13	47					
T15	DATA BUS 14	43					
T16	DATA BUS 15	42		1100			
T17	DISCRETE 0	35		750			
T18	DISCRETE 1	81					
T19	DISCRETE 2	84					
T20	DISCRETE 3	32					
T21	DISCRETE 4	78	0	750	mV	1-4	12,13

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* TABLE XX. Output crosstalk, high level, test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes	Figs
T22	DISCRETE 5	28	0	750	mV	1-4	12,13
T23	DISCRETE 6	75					
T24	DISCRETE 7	24					
T25	-DRVR ENBL 0	82					
T26	-DRVR ENBL 1	86					
T27	-DRVR ENBL 2	37					
T28	-DRVR ENBL 3	36					
T29	-DRVR ENBL 4	73					
T30	-DRVR ENBL 5	79					
T31	-DRVR ENBL 6	77					
T32	-DRVR ENBL 7	29		750			
T33	-IE CMD REQ	70		400			
T34	-IE DATA REQ	69					
T35	-OE REQ	19					
T36	DIFF A0 OUT	16					
T37	DIFF B0 OUT	64					
T38	DIFF A1 OUT	14					
T39	DIFF B1 OUT	63					
T40	-RCVR 0 ENBL	59					
T41	-RCVR 1 ENBL	62					
T42	-RECV ENBL	8					
T43	-LINK NOT BSY	60	0	400	mV	1-4	12,13

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* TABLE XX. Output crosstalk, high level, test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes	Figs
T44	-DREQ	9	0	400	mV	1-4	12,13
T45	-DATA XFR STB	12					
T46	SEND CLOCK	54					
T47	-DRVR 1 DATA OUT	52					
T48	-BIPLR 1 OUT	11					
T49	-DRVR DATA OUT	55					
T50	-RCVR 1 DATA	65					
T51	-RCVR DATA	66					
T52	+BIAS	27					
T53	PULLUP 1 (1K)	22					
T54	PULLUP 2 (1K)	72	0	400	mV	1-4	12,13

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NOTES:

*1. Output crosstalk, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.0 V	± 0.25 V	T1-T54
Power supply current limit	1.058 A	NA	T1-T54
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 T1-T54 T1-T54
Output crosstalk measurement	NA	± 20 mV	T1-T54

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) Use probe 'A' to verify the pattern voltage and pattern repetition rate of 1000 nS ± 200 nS, 50 percent duty cycle.

(d) The module inputs shall not violate the minimum input timing requirements shown on figure 8.

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NOTES: - Continued.

(e) For each output under test in table XX, connect probe 'B' and visually inspect the output for the most negative spike as indicated on figure 13. The test circuit shall be as indicated on figure 12.

(f) The maximum probe capacitance is 10 pF.

3. All outputs shall be loaded as specified on figure 7.

4. Abbreviations used are defined in 6.1.

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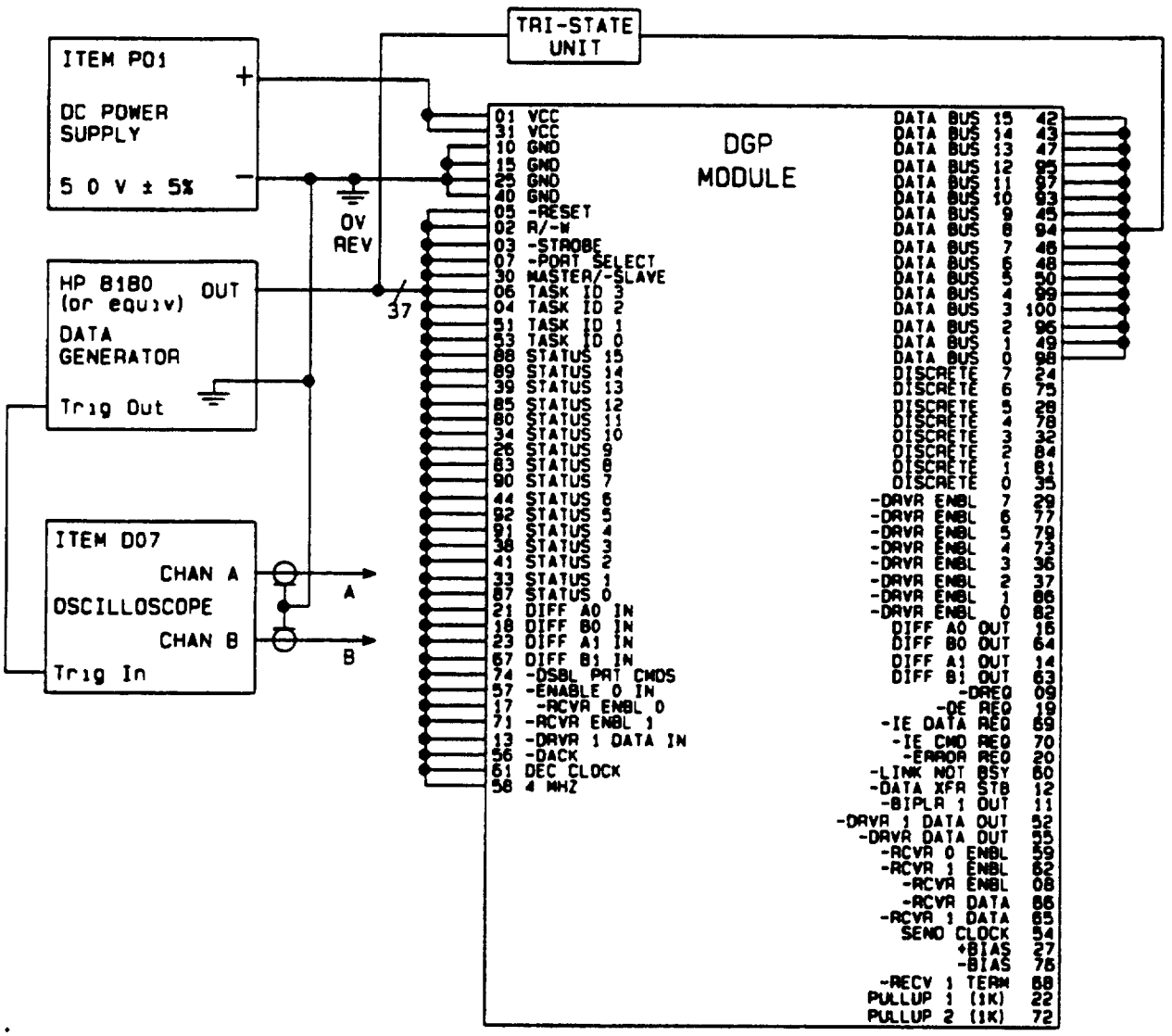


FIGURE 12. Output crosstalk, high and low level, test circuit for tables XX and XXI.

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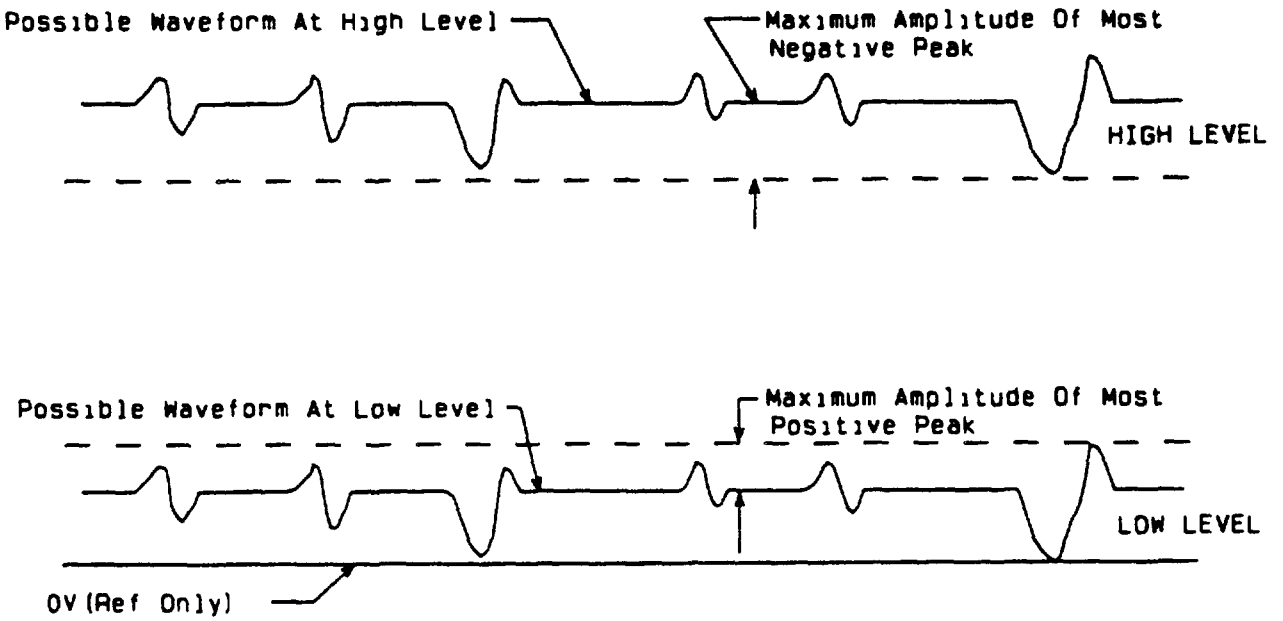


FIGURE 13. Output crosstalk, high and low level, test waveforms for tables XX and XXI.

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* TABLE XXI. Output crosstalk, low level, test.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes	Figs
U1	DATA BUS 0	98	0	1100	mV	1-4	12,13
U2	DATA BUS 1	49					
U3	DATA BUS 2	96					
U4	DATA BUS 3	100					
U5	DATA BUS 4	99					
U6	DATA BUS 5	50					
U7	DATA BUS 6	48					
U8	DATA BUS 7	46					
U9	DATA BUS 8	94					
U10	DATA BUS 9	45					
U11	DATA BUS 10	93					
U12	DATA BUS 11	97					
U13	DATA BUS 12	95					
U14	DATA BUS 13	47					
U15	DATA BUS 14	43					
U16	DATA BUS 15	42		1100			
U17	DISCRETE 0	35		750			
U18	DISCRETE 1	81					
U19	DISCRETE 2	84					
U20	DISCRETE 3	32					
U21	DISCRETE 4	78	0	750	mV	1-4	12,13

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* TABLE XXI. Output crosstalk, low level, test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes	Figs
U22	DISCRETE 5	28	0	750	mV	1-4	12,13
U23	DISCRETE 6	75					
U24	DISCRETE 7	24					
U25	-DRVR ENBL 0	82					
U26	-DRVR ENBL 1	86					
U27	-DRVR ENBL 2	37					
U28	-DRVR ENBL 3	36					
U29	-DRVR ENBL 4	73					
U30	-DRVR ENBL 5	79					
U31	-DRVR ENBL 6	77					
U32	-DRVR ENBL 7	29		750			
U33	-IE CMD REQ	70		400			
U34	-IE DATA REQ	69					
U35	-OE REQ	19					
U36	DIFF A0 OUT	16					
U37	DIFF B0 OUT	64					
U38	DIFF A1 OUT	14					
U39	DIFF B1 OUT	63					
U40	-RCVR 0 ENBL	59					
U41	-RCVR 1 ENBL	62					
U42	-RCVR ENBL	8	0	400	mV	1-4	12,13

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* TABLE XXI. Output crosstalk, low level, test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes	Figs
U43	-LINK NOT BSY	60	0	400	mV	1-4	12,13
U44	-DREQ	9					
U45	-DATA XFR STB	12					
U46	SEND CLOCK	54					
U47	-DRVR 1 DATA OUT	52					
U48	-BIPLR 1 OUT	11					
U49	-DRVR DATA OUT	55					
U50	-RCVR 1 DATA	65					
U51	-RCVR DATA	66					
U52	-BIAS	76	0	400	mV	1-4	12,13

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NOTES:

*1. Output crosstalk, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.0 V	± 0.25 V	U1-U52
Power supply current limit	1.058 A	NA	U1-U52
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 U1-U52 U1-U52
Output crosstalk measurement	NA	± 20 mV	U1-U52

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) Use probe 'A' to verify the pattern voltage and pattern repetition rate of 1000 nS ± 200 nS, 50 percent duty cycle.

* (d) For each output under test in table XXI, connect probe 'B' and visually inspect the output for the most positive spike as indicated on figure 13. The test circuit shall be as indicated on figure 12. The module inputs shall not violate the minimum input timing requirements shown on figure 8.

(e) The maximum probe capacitance is 10 pF.

3. All outputs shall be loaded as specified on figure 7.

4. Abbreviations used are defined in 6.1.

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* TABLE XXII. Crosstalk duration, high and low level.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes	Fig.
V1	DATA BUS 0	98	0	75	nS	1-4	14
V2	DATA BUS 1	49					
V3	DATA BUS 2	96					
V4	DATA BUS 3	100					
V5	DATA BUS 4	99					
V6	DATA BUS 5	50					
V7	DATA BUS 6	48					
V8	DATA BUS 7	46					
V9	DATA BUS 8	94					
V10	DATA BUS 9	45					
V11	DATA BUS 10	93					
V12	DATA BUS 11	97					
V13	DATA BUS 12	95					
V14	DATA BUS 13	47					
V15	DATA BUS 14	43					
V16	DATA BUS 15	42		75			
V17	DISCRETE 0	35		153			
V18	DISCRETE 1	81					
V19	DISCRETE 2	84					
V20	DISCRETE 3	32					
V21	DISCRETE 4	78					
V22	DISCRETE 5	28	0	153	nS	1-4	14

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* TABLE XXII. Crosstalk duration, high and low level - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes	Fig.
V23	DISCRETE 6	75	0	153	nS	1-4	14
V24	DISCRETE 7	24		153			
V25	-DRVR ENBL 0	82		128			
V26	-DRVR ENBL 1	86					
V27	-DRVR ENBL 2	37					
V28	-DRVR ENBL 3	36					
V29	-DRVR ENBL 4	73					
V30	-DRVR ENBL 5	79					
V31	-DRVR ENBL 6	77					
V32	-DRVR ENBL 7	29		128			
V33	DATA BUS 0	98		75			
V34	DATA BUS 1	49					
V35	DATA BUS 2	96					
V36	DATA BUS 3	100					
V37	DATA BUS 4	99					
V38	DATA BUS 5	50					
V39	DATA BUS 6	48					
V40	DATA BUS 7	46					
V41	DATA BUS 8	94					
V42	DATA BUS 9	45					
V43	DATA BUS 10	93	0	75	nS	1-4	14

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* TABLE XXII. Crosstalk duration, high and low level - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes	Fig.
V44	DATA BUS 11	97	0	75	nS	1-4	14
V45	DATA BUS 12	95					
V46	DATA BUS 13	47					
V47	DATA BUS 14	43					
V48	DATA BUS 15	42		75			
V49	DISCRETE 0	35		153			
V50	DISCRETE 1	81					
V51	DISCRETE 2	84					
V52	DISCRETE 3	32					
V53	DISCRETE 4	78					
V54	DISCRETE 5	28					
V55	DISCRETE 6	75					
V56	DISCRETE 7	24		153			
V57	-DRVR ENBL 0	82		128			
V58	-DRVR ENBL 1	86					
V59	-DRVR ENBL 2	37					
V60	-DRVR ENBL 3	36					
V61	-DRVR ENBL 4	73					
V62	-DRVR ENBL 5	79					
V63	-DRVR ENBL 6	77					
V64	-DRVR ENBL 7	29	0	128	nS	1-4	14

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NOTES:

1. Output crosstalk duration test tolerance table.

Condition	Value	Accuracy	Current limit	Test cond
Power supply	5.0 V	± 50 mV	935 mA	V1-V64
Pattern voltage				
High (VH)	3.0 V	± 100 mV	NA	V1-V64
Low (VL)	0.0 V	± 100 mV	NA	V1-V64
Crosstalk duration measurement	NA	± .825 nS	NA	V1-V16 V33-V48
		± 1.63 nS		V17-V24 V49-V56
	NA	± 1.4 nS	NA	V25-V32 V57-V64

2. The following procedure is required to perform this test.

(a) Apply power to the module. The module shall have the power supply voltage applied to all Vcc pins. The DC return of the power supply shall be connected to all ground pins of the module. The power supply voltage overshoot shall not exceed 1.5 V.

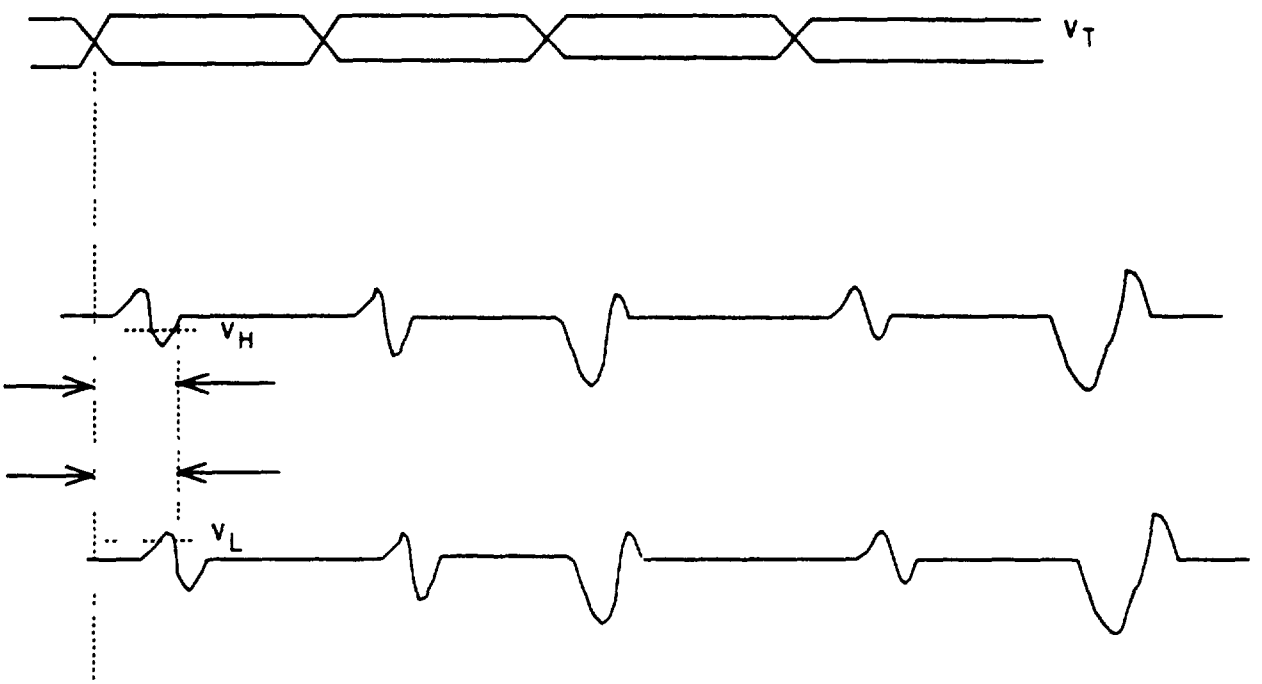
(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line. The module inputs shall not violate the minimum input timing requirements shown on figure 8.

(c) The duration shall be measured from the triggering edge of the input, as shown on figure 14.

3. All output pins shall be loaded with the appropriate load as shown on figure 7, except C = 50 ± 5 pF.

4. Abbreviation used are defined in 6.1.

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NOTES:

1. VT = 1.3 V

2. VH = VL = 400 mV

3. The crosstalk duration is measured from the triggering edge of the input to the point where output crosstalk returns to the specified voltage. If the crosstalk amplitude is smaller than the duration measurement voltage (VH or VL) the crosstalk duration shall be recorded as 0 nS.

FIGURE 14. Crosstalk duration, high and low level, test.

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TABLE XXIII. Input capacitance test.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
W1	STATUS 0	87	2	25	pF	1-3
W2	STATUS 1	33		35		
W3	STATUS 2	41		25		
W4	STATUS 3	38				
W5	STATUS 4	91				
W6	STATUS 5	92				
W7	STATUS 6	44				
W8	STATUS 7	90		25		
W9	STATUS 8	83		40		
W10	STATUS 9	26				
W11	STATUS 10	34				
W12	STATUS 11	80		40		
W13	STATUS 12	85		35		
W14	STATUS 13	39				
W15	STATUS 14	89				
W16	STATUS 15	88				
W17	-RESET	5				
W18	TASK ID 0	53		35		
W19	TASK ID 1	51		25		
W20	TASK ID 2	4				
W21	TASK ID 3	6	2	25	pF	1-3

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TABLE XXIII. Input capacitance test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
W22	-STROBE	3	2	25	pF	1-3
W23	R/-W	2				
W24	-DACK	56				
W25	-PORT SELECT	7		25		
W26	DEC CLOCK	61		40		
W27	4 MHZ	58		40		
W28	DIFF A0 IN	21		75		
W29	DIFF B0 IN	18				
W30	DIFF A1 IN	23				
W31	DIFF B1 IN	67		75		
W32	-RCVR ENBL 0	17		35		
W33	-RCVR ENBL 1	71				
W34	-DRVR 1 DATA IN	13		35		
W35	-ENABLE 0 IN	57		25		
W36	MASTER/-SLAVE	30		30		
W37	-DSBL PRT CMDS	74				
W38	DATA BUS 0	98				
W39	DATA BUS 1	49				
W40	DATA BUS 2	96				
W41	DATA BUS 3	100				
W42	DATA BUS 4	99	2	30	pF	1-3

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TABLE XXIII. Input capacitance test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
W43	DATA BUS 5	50	2	30	pF	1-3
W44	DATA BUS 6	48				
W45	DATA BUS 7	46		30		
W46	DATA BUS 8	94		40		
W47	DATA BUS 9	45				
W48	DATA BUS 10	93				
W49	DATA BUS 11	97				
W50	DATA BUS 12	95				
W51	DATA BUS 13	47				
W52	DATA BUS 14	43				
W53	DATA BUS 15	42	2	40	pF	1-3

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NOTES:

1. Input capacitance test tolerance table.

Condition	Value	Accuracy	Test condition
Capacitance meter frequency	1 MHz	± 100 KHz	W1-W53
Capacitance meter voltage	15 mV	± 1.5 mV	W1-W53
Capacitance measurement	NA | NA	± 1.15 pF ± 2.15 pF ± 1.15 pF	W1-W27 W28,W29 W30-W53

2. The capacitance meter used in this test shall be item C05. The negative terminal shall be connected to the module ground and the positive terminal to the pin indicated in table XXIII.

3. Abbreviations used are defined in 6.1.

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TABLE XXIV. Output capacitance test.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
X1	-IE CMD REQ	70	2	25	pF	1-3
X2	-IE DATA REQ	69				
X3	-OE REQ	19				
X4	-ERROR REQ	20		25		
X5	-RCVR DATA	66		50		
X6	-RCVR 1 DATA	65	2	25	pF	1-3

NOTES:

1. Output capacitance test tolerance table.

Condition	Value	Accuracy	Test condition
Capacitance meter frequency	1 MHz	± 100 KHz	X1-X6
Capacitance meter voltage	15 mV	± 1.5 mV	X1-X6
Capacitance measurement	NA | NA	± 1.15 pF ± 2.4 pF ± 1.15 pF	X1-X4 X5 X6

2. The capacitance meter used in this test shall be item C05. The negative terminal shall be connected to the module ground and the positive terminal to the pin indicated in table XXIV.

3. Abbreviations used are defined in 6.1.

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* TABLE XXV. Threshold test.

Test cond	Parameter	Notes	Fig
Y1	Threshold	1-4	8
Y2	Threshold	1-4	8

NOTES:

*1. Threshold test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V 5.5 V	± 45 mV ± 55 mV	Y1 Y2
Power supply current limit	1.058 A	NA	Y1,Y2
Pattern voltage (See note 3) high, (VH) high, (VH) low, (VL) low, (VL)	 2.2 V 2.0 V 0.8 V 0.7 V	 ± 20 mV ± 7 mV ± 8 mV ± 22 mV	 Y1,Y2 Y1,Y2 Y1,Y2 Y1,Y2
Output voltage measurement high, (Voh) high, (Voh) low, (Vol)	 NA | NA	 ± 105 mV ± 100 mV ± 25 mV	 Y1,Y2 Y1,Y2 Y1,Y2

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

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NOTES: - Continued.

(b) Precondition the module under test and verify the output patterns by monitoring the outputs and comparing the results to the output low and high levels specified in the output voltage tests. The module inputs shall not violate the minimum input timing requirements shown on figure 8.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

3. The high and low level input threshold voltage for each pin is as specified in 6.3.1.

4. Abbreviations used are defined in 6.1.

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* TABLE XXVI. Disable time test.

Test cond	Signal name	C to pin	25 ± 5°C Initial limits Min	Max	Units	Figs
----------	Low to high impedance (tPLZ)	----------				
Z1	-DACK to DATA BUS 0	98	4	260	nS	15,16
Z2	-DACK to DATA BUS 1	49				
Z3	-DACK to DATA BUS 2	96				
Z4	-DACK to DATA BUS 3	100				
Z5	-DACK to DATA BUS 4	99				
Z6	-DACK to DATA BUS 5	50				
Z7	-DACK to DATA BUS 6	48				
Z8	-DACK to DATA BUS 7	46				
Z9	-DACK to DATA BUS 8	94				
Z10	-DACK to DATA BUS 9	45				
Z11	-DACK to DATA BUS 10	93				
Z12	-DACK to DATA BUS 11	97				
Z13	-DACK to DATA BUS 12	95				
Z14	-DACK to DATA BUS 13	47				
Z15	-DACK to DATA BUS 14	43				
Z16	-DACK to DATA BUS 15	42		260		
Z17	-STROBE to DATA BUS 0	98		130		
Z18	-STROBE to DATA BUS 1	49				
Z19	-STROBE to DATA BUS 2	96				
Z20	-STROBE to DATA BUS 3	100	4	130	nS	15,16

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* TABLE XXVI. Disable time test - Continued.

Test cond	Signal name	C to pin	25 ± 5°C Initial limits Min	Max	Units	Figs
---------- Low to high impedance(tPLZ) ----------						
Z21	-STROBE to DATA BUS 4	99	4	130	nS	15,16
Z22	-STROBE to DATA BUS 5	50				
Z23	-STROBE to DATA BUS 6	48				
Z24	-STROBE to DATA BUS 7	46				
Z25	-STROBE to DATA BUS 8	94				
Z26	-STROBE to DATA BUS 9	45				
Z27	-STROBE to DATA BUS 10	93				
Z28	-STROBE to DATA BUS 11	97				
Z29	-STROBE to DATA BUS 12	95				
Z30	-STROBE to DATA BUS 13	47				
Z31	-STROBE to DATA BUS 14	43				
Z32	-STROBE to DATA BUS 15	42		130		
High to high impedance (tPHZ)						
Z33	-DACK to DATA BUS 0	98		260		
Z34	-DACK to DATA BUS 1	49				
Z35	-DACK to DATA BUS 2	96				
Z36	-DACK to DATA BUS 3	100				
Z37	-DACK to DATA BUS 4	99				
Z38	-DACK to DATA BUS 5	50				
Z39	-DACK to DATA BUS 6	48	4	260	nS	15,16

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* TABLE XXVI. Disable time test - Continued.

Test cond	Signal name	C to pin	25 ± 5°C Initial limits Min	Max	Units	Figs
---------- High to high impedance (tPHZ) ----------						
Z40	-DACK to DATA BUS 7	46	4	260	nS	15,16
Z41	-DACK to DATA BUS 8	94				
Z42	-DACK to DATA BUS 9	45				
Z43	-DACK to DATA BUS 10	93				
Z44	-DACK to DATA BUS 11	97				
Z45	-DACK to DATA BUS 12	95				
Z46	-DACK to DATA BUS 13	47				
Z47	-DACK to DATA BUS 14	43				
Z48	-DACK to DATA BUS 15	42		260		
Z49	-STROBE to DATA BUS 0	98		130		
Z50	-STROBE to DATA BUS 1	49				
Z51	-STROBE to DATA BUS 2	96				
Z52	-STROBE to DATA BUS 3	100				
Z53	-STROBE to DATA BUS 4	99				
Z54	-STROBE to DATA BUS 5	50				
Z55	-STROBE to DATA BUS 6	48				
Z56	-STROBE to DATA BUS 7	46				
Z57	-STROBE to DATA BUS 8	94				
Z58	-STROBE to DATA BUS 9	45				
Z59	-STROBE to DATA BUS 10	93	4	130	nS	15,16

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* TABLE XXVI. Disable time test - Continued.

Test cond	Signal name	C to pin	25 ± 5°C Initial limits Min	Max	Units	Figs
---------- High to high impedance (tPHZ) ----------						
Z60	-STROBE to DATA BUS 11	97	4	130	nS	15,16
Z61	-STROBE to DATA BUS 12	95				
Z62	-STROBE to DATA BUS 13	47				
Z63	-STROBE to DATA BUS 14	43				
Z64	-STROBE to DATA BUS 15	42	4	130	nS	15,16

NOTES:

*1. Output disable time test tolerance table.

Condition	Value	Accuracy	Current limit	Test cond
Power supply	5.0 V	± 0.25 V	925 mA	Z1-Z64
Pattern voltage				
High, (VH)	3.0 V	± 100 mV	NA	Z1-Z64
Low, (VL)	0.0 V	± 100 mV	NA	Z1-Z64
Disable time measurement	NA	± 12.8 nS	NA	Z1-Z16, Z33-Z48
	NA	± 6.3 nS	NA	Z17-Z32, Z49-Z64

*2. The following procedure is required to perform this test.

(a) Apply power to the module. The module shall have the power supply voltage applied to all Vcc pins. The DC return of the power supply shall be connected to all ground pins of the module. The power supply voltage overshoot shall not exceed 2.0 V.

(b) The module inputs shall not violate the minimum input timing requirements shown on figure 8.

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NOTES: - Continued.

(c) Use probe 'A' to verify the pattern voltage and pattern repetition rate of 1000 nS ± 200 nS, 50 percent duty cycle.

(d) The maximum probe capacitance is 10 pF.

3. All outputs shall be loaded as specified on figure 7 with CL = 50 pF. Signals DATA BUS 0-15 are bidirectional and will be loaded as shown on figure 8 when programmed to the outputs.

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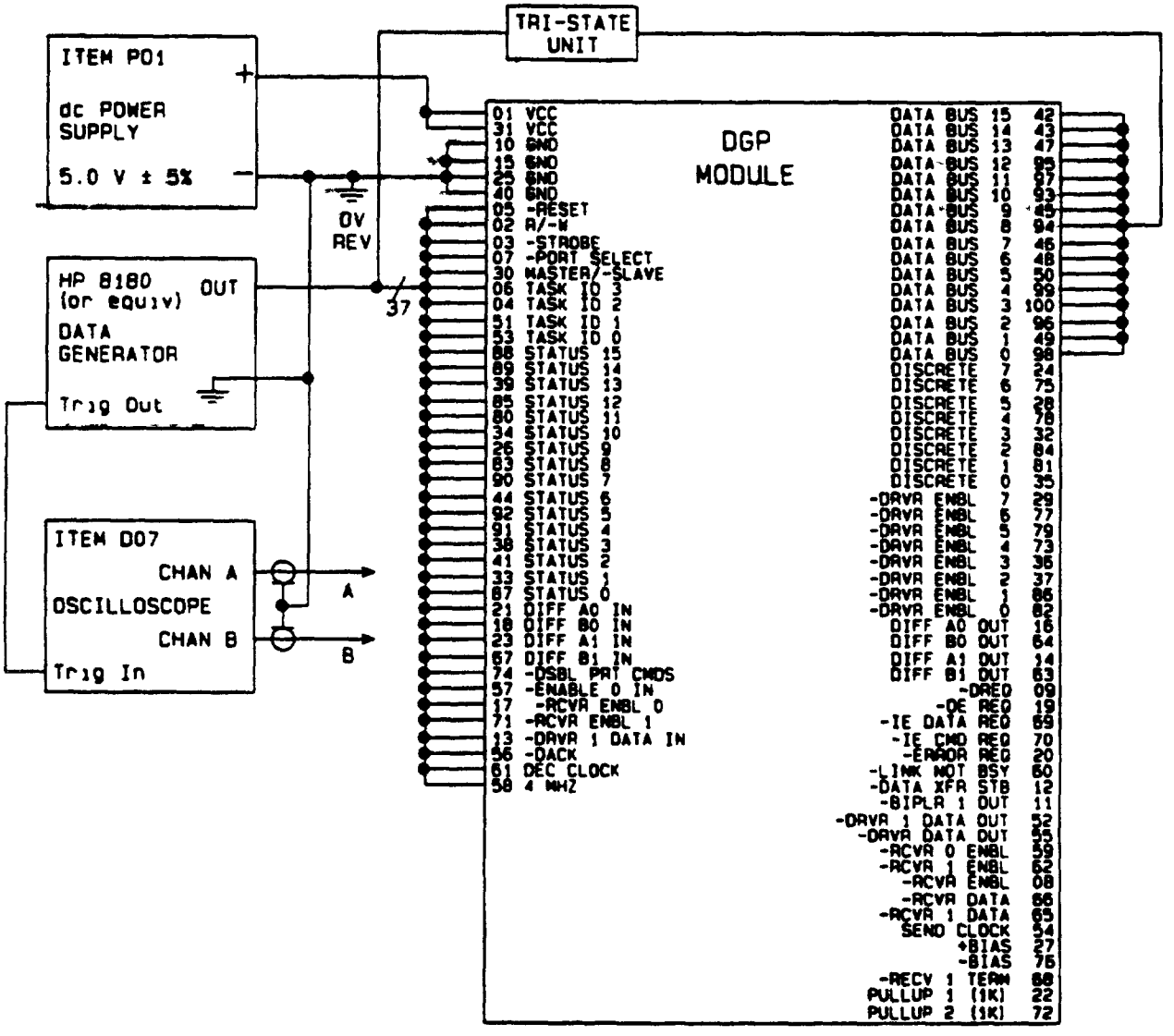


FIGURE 15. Disable time test circuit for table XXVI.

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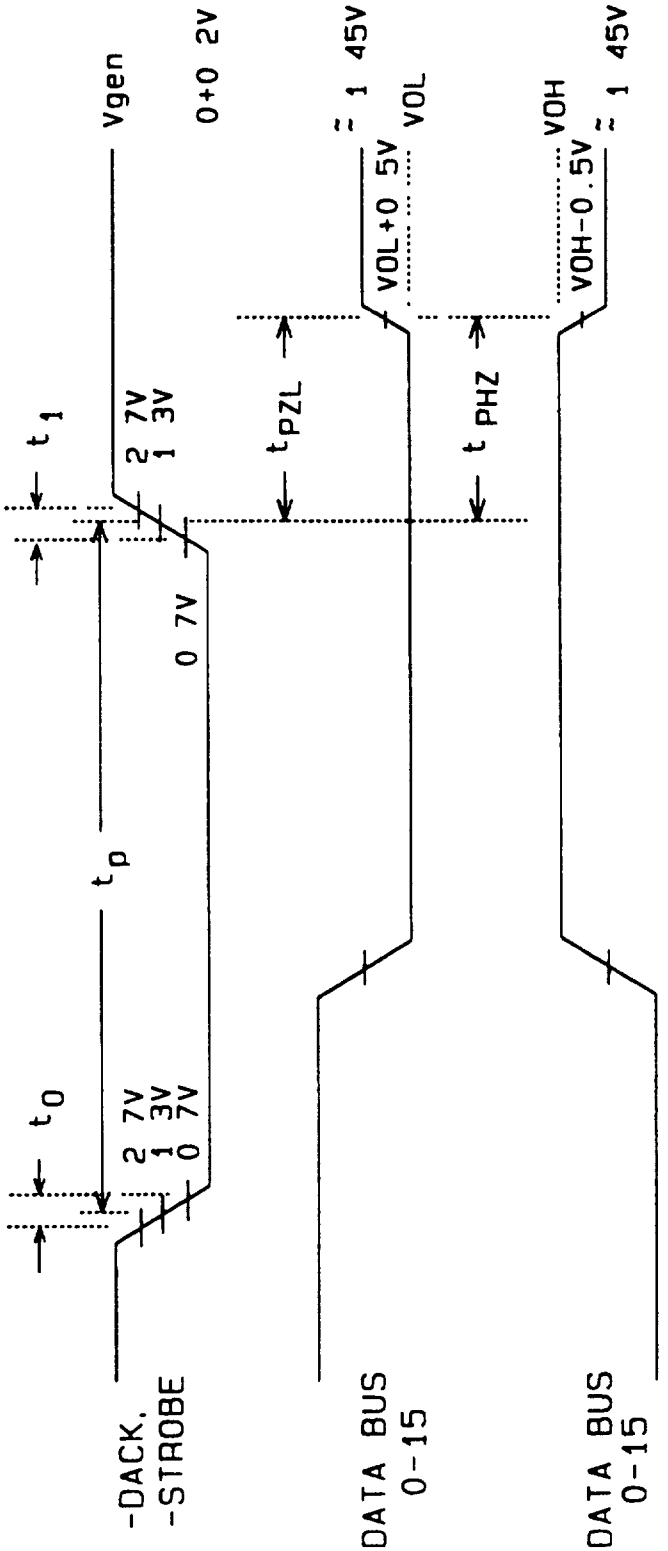


FIGURE 16. Disable time test waveforms for table XXVI.

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* TABLE XXVII. Noise margin test.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits		Units	Notes	Figs
			Min	Max			
AA1	-ERROR REQ	20	0	800	mV	1-4	17, 18

NOTES:

*1. Noise margin test tolerance table.

Condition	Value	Accuracy	Current limits	Test cond
Power supply	5.5 V	± 0.275 V	925 mA	AA1
Pattern voltage High (VH) Low (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 NA NA	 AA1 AA1
Noise margin measurement	NA	± 40 mV	NA	AA1

*2. The following procedure is required to perform this test.

(a) Apply power to the module. The module shall have the power supply voltage applied to all Vcc pins. The DC return of the power supply shall be connected to all ground pins of the module. The power supply voltage overshoot shall not exceed 2.0 V.

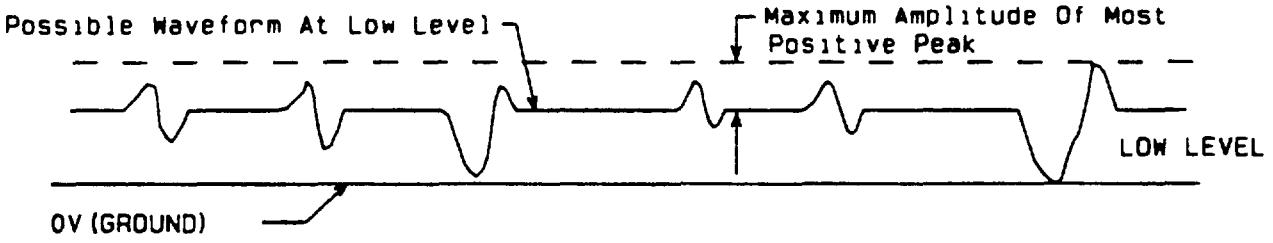
(b) Use probe 'A' to verify the pattern voltage and pattern repetition rate of 1000 nS ± 200 nS, 50 percent duty cycle.

(c) The maximum probe capacitance is 10 pF.

*3. All outputs shall be loaded as specified on figure 7 with CL = 50 pF.

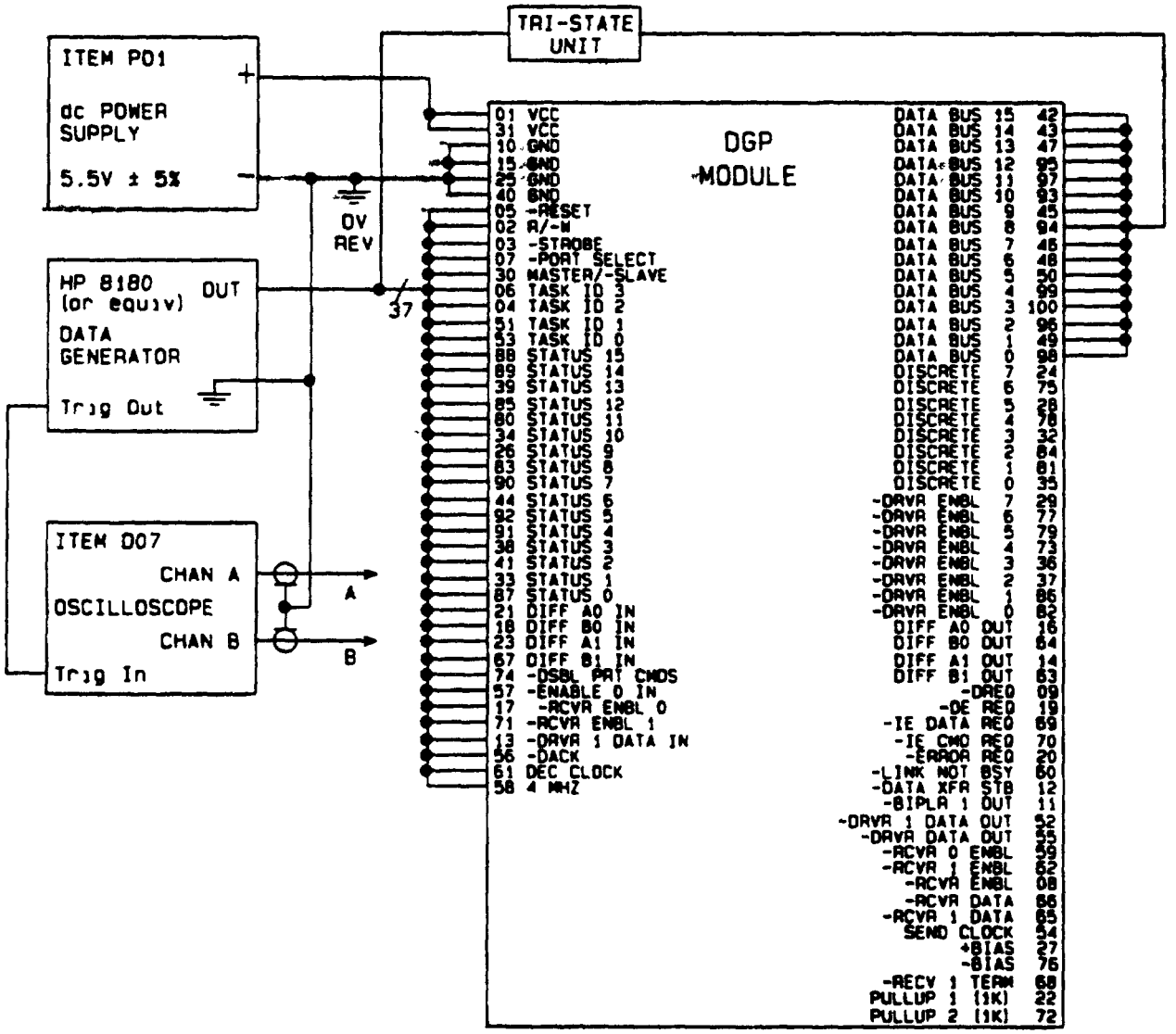
*4. Abbreviations used are defined in 6.1.

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* FIGURE 17. Noise margin test circuit waveforms for table XXVII.

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* FIGURE 18. Noise margin test circuit for table XXVII.

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3 DOCUMENT TITLE
Modules, Digital, Standard Electronic, Interface Module, Key Code DGP

4 NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

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